Florida Artificial Reef Estuarine Deployment Footprints with Cost: Background and supporting documentation

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Associated data file: FL_AR_Estuarine_Footprints_with_Cost-DATA(2026-06-23).xlsx

Background

Encompassing 34 of 35 different coastal counties spread along 1,357 miles of ocean fronting coastline, Florida manages one of the most diverse, and most active artificial reef programs in the United States. As of October 2025, the FWC Artificial Reef Program reports over 4,476 artificial reef deployment locations state-wide. Artificial Reefs are about evenly divided between state and federal waters and range from 1 foot in depth to 511 feet (average depth 68 feet). Of the 4,476 artificial reef deployments, a total of 321 are located within Florida estuaries. This report documents the details of those estuarine deployments, including location, estimated footprint, cost and other parameters. Example footprint calculations and associated figures are encompassed within this document.

Data Access and Use Constraints

These data are available without restriction. It is strongly recommended that this data is directly acquired from FWC and not indirectly through other sources which may have changed the data in some way. FWC makes no claims as to the data's suitability for other purposes. Acknowledgement of the FWC Division of Marine Fisheries as the data source and originator, would be appreciated in any products developed from these data, and such acknowledgment as is standard for citation and legal practices for data source is expected by users of this data. Please cite the original metadata when using portions of the record to create a similar record of slightly altered data, such as a re-projection. If any data are modified or adjusted, please share the edited information with FWC. Users should be aware that comparison with other data sets for the same area from other time periods may be inaccurate due to inconsistencies resulting from changes in mapping conventions, data collection, and computer processes over time. FWC shall not be liable for improper or incorrect use of this data. These data are not legal documents and are not to be used as such. This is not a survey data set and should not be utilized as such. These data are not to be used for navigation.

Location Disclaimer

Please note that reef materials can move, degrade, or become buried over time. This is especially true for metal materials with thicknesses less than ¼" or lightweight materials. Additionally, many reef locations in the database are based on historical data with inherent potential location inaccuracies that may not have been ground-truthed since their deployment. If you have updated information regarding the condition or location of an artificial reef please contact the FWC Artificial Reef Database Administrator via email at artificialreefdeployments@myfwc.com.

Other Download Sources

The complete list of Florida artificial reef locations are also available as an excel spreadsheet on the artificial reef section's page: http://myfwc.com/conservation/saltwater/artificial-reefs/locate-reefs/

For an interactive map as well as downloads in KML and ESRI Shapefile formats, please visit FWC's Florida Fish and Wildlife Research Institute GIS & Mapping Data Download page: http://geodata.myfwc.com/datasets/

Data Field Definitions (Metadata)

DeployID - A unique six-digit alphanumeric identifier for each deployment record, assigned by FWC.

DeployDate - The date of deployment (YYYYMMDD). Double zeros indicate estimates based on the best available historic information.

Description - Information relating to the materials placed during the artificial reef deployment.

Tonnage - Dry weight of the reef material. Use this field to judge the approximate overall size of the reef.

Area - Footprint of the artificial reef materials on the bottom measured in square feet.

Depth - Depth (in feet) to the bottom at the time of deployment.

Relief - Height (in feet) of the reef materials off the bottom.

Deployment Name - The common name for the deployment site.

Material Type - The type of reef material that makes up the majority of the deployment, prefixed by six general categories (CONCRETE, METAL, PLASTIC, ROCK, RUBBER and SHELL).

Structure Category- The material shape and items deployed that are composed from each material type. Pre-fabricated reef modules and marine vessels are labelled as 'MODULE' or 'VESSEL' regardless of material composition.

Permit Area **Name** – The name of the permitted area each reef is deployed in (if available).

Coast – The coastline each reef is deployed off of (Gulf of America or Atlantic Ocean).

County - The Florida County jurisdiction where the artificial reef project is located.

Coordinate Accuracy (Location) - A value of *High*, *Medium*, or *Low* assigned based on the method of how the deployment coordinates were acquired. The FWC Artificial Reef Program continuously strives to improve coordinate information about

each deployment location. If you have updated information regarding the condition or location of an artificial reef please contact the FWC Artificial Reef Database Administrator via email at artificialreefdeployments@myfwc.com.

High location accuracy represents deployment locations that were acquired using modern GPS systems such as WAAS or DGPS.

Medium location accuracy represents deployments that were identified using GPS systems with lower accuracy compared to modern units; these locations have also not been verified.

Low location accuracy represents historical deployments recorded with LORAN coordinates and later converted using computer algorithms to degrees decimal minutes and not yet verified using a modern GPS system. These deployments may have a large range of spatial error depending on the LORAN conversion.

Permit Area/ SS in sq ft – The entire permit area or entire area of a sidescan survey (measured in sq ft) that were used to estimate the area of an artificial reef deployment.

Permit Area/ SS used in estimate – The portion of the permit area or sidescan survey that contained artificial reef materials and were used to estimate the area of an artificial reef deployment.

Total Tonnage Used for Calculations – The total tonnage of all deployments within a permitted area or side scan survey that were used when estimating the area of an artificial reef deployment.

Footprint Estimation Description – The description of how the area was estimated for any deployments where the area was previously unknown but information was available to calculate an estimate of the materials.

Latitude (DD) - Latitude in decimal degrees.

Longitude (DD) - Longitude in decimal degrees.

Funding Source - Category – Whether the funding source was from the federal, state, or local government, or was a private donation (non-governmental funds).

Funding Source - Entity – The name of the entity that served as a funding source for each artificial reef deployment.

Funding Source - Fund – The name of the fund applied to each funding source for each artificial reef deployment.

FWC Grant –Grant number assigned by the Florida Fish and Wildlife Conservtaion Commission (FWC) for artificial reef projects that received funding from an FWC grant. Multiple DeployIDs may be affiliated with one FWC Grant.

Federal Grant – Federal grant number for artificial reef projects that received federal funding from an FWC grant. Multiple FWC Grants may be affiliated with one Federal Grant.

Local Govt Funding (byDeployID) - Funding from local governments (county or city municipalities) for each artificial reef deployment.

State Funding (byDeployID) - Funding from a state source for each artificial reef deployment.

Federal Funding (byDeployID) - Funding from a federal source for each artificial reef deployment.

Other Funding - Funding donated from a private or unknown source for each artificial reef deployment.

Total Funding(byDeployID) – The total funding from all funding sources for each artificial reef deployment.

Deployment cost by tonnage – The cost for each artificial reef deployment based on tonnage. For projects where multiple deployments are affiliated with one FWC Grant the cost was split based on deployment tonnage.

Cost per ton – The total project cost of an FWC Grant divided by the total tonnage of the deployments affiliated within each FWC Grant.

Footprint Estimation Methodology:

Methodology for estimation of artificial reef footprints varied for each individual reef based on information that was available to the program. Information on estuarine reefs was extrapolated from the Florida Artificial Reef Database (FARD) to focus on reefs that fell within the FDEP Numeric Nutrient Criteria (NNC) Estuary Nutrient Regions GIS layer. From this dataset, 321 artificial reefs were classified as estuarine reefs, and 191 of the reefs had an unknown deployment area (footprint). The deployment footprints were estimated depending on the individual deployment and any known information that could be taken from the Material Placement Reports (MPR), tonnage, description, side scan information, permit size, or bathymetry. The description for each estimate is within the Estimation Description column of the Estuarine Deployment data sheet. For estimates based on side scan or permit areas, the information and imagery is included below.

Duval County Deployments:

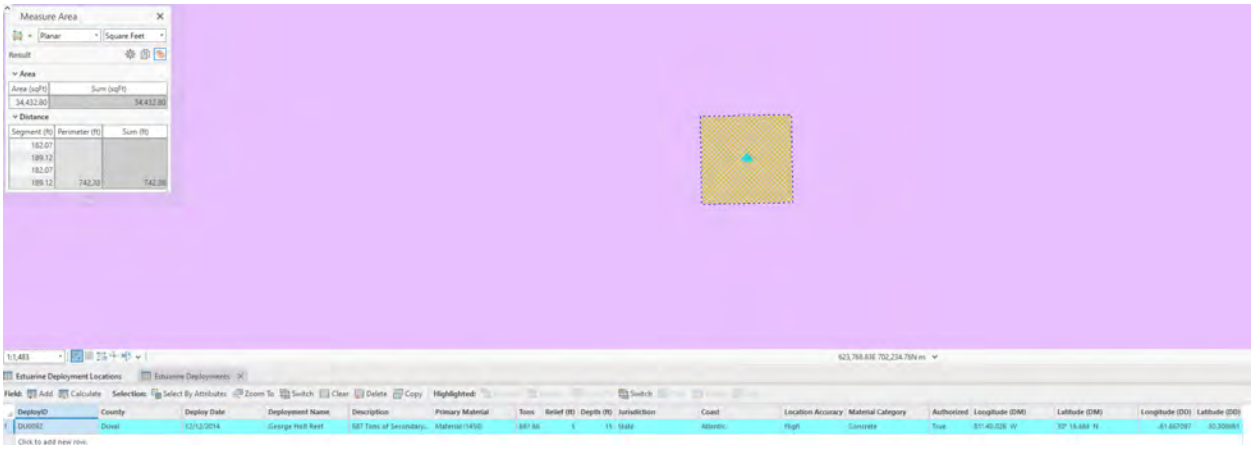


Fig 1. DU0092 within 34,432.80 sq ft George Holt Reef permitted area.

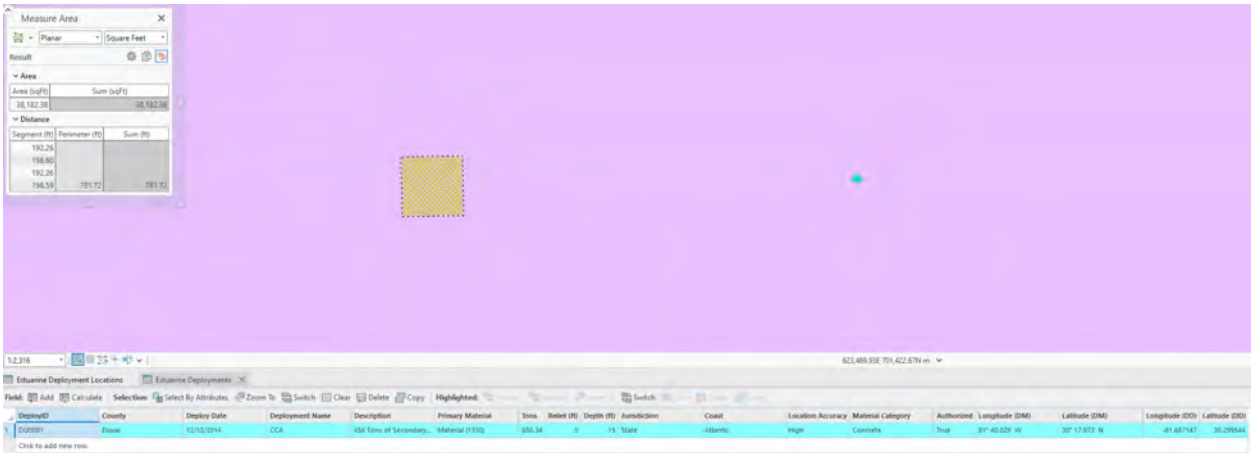


Fig 2. DU0091 outside of 38,182.38 sq ft CCA permitted area.

Escambia County Deployments:

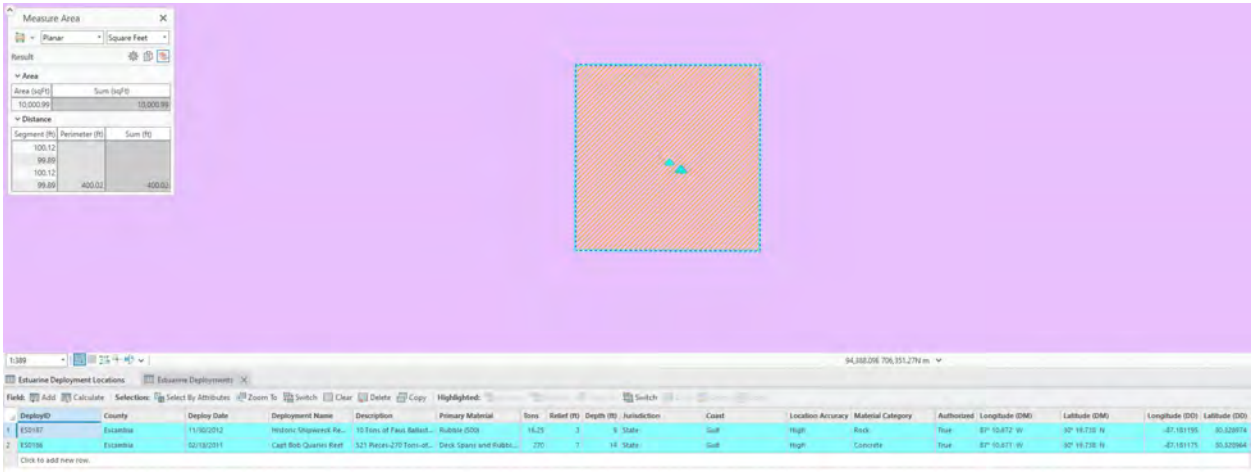


Fig 3. ES0186 and ES0187 inside 10,000 sq ft Escambia Inshore-North permitted area.

Hillsborough County Deployments:

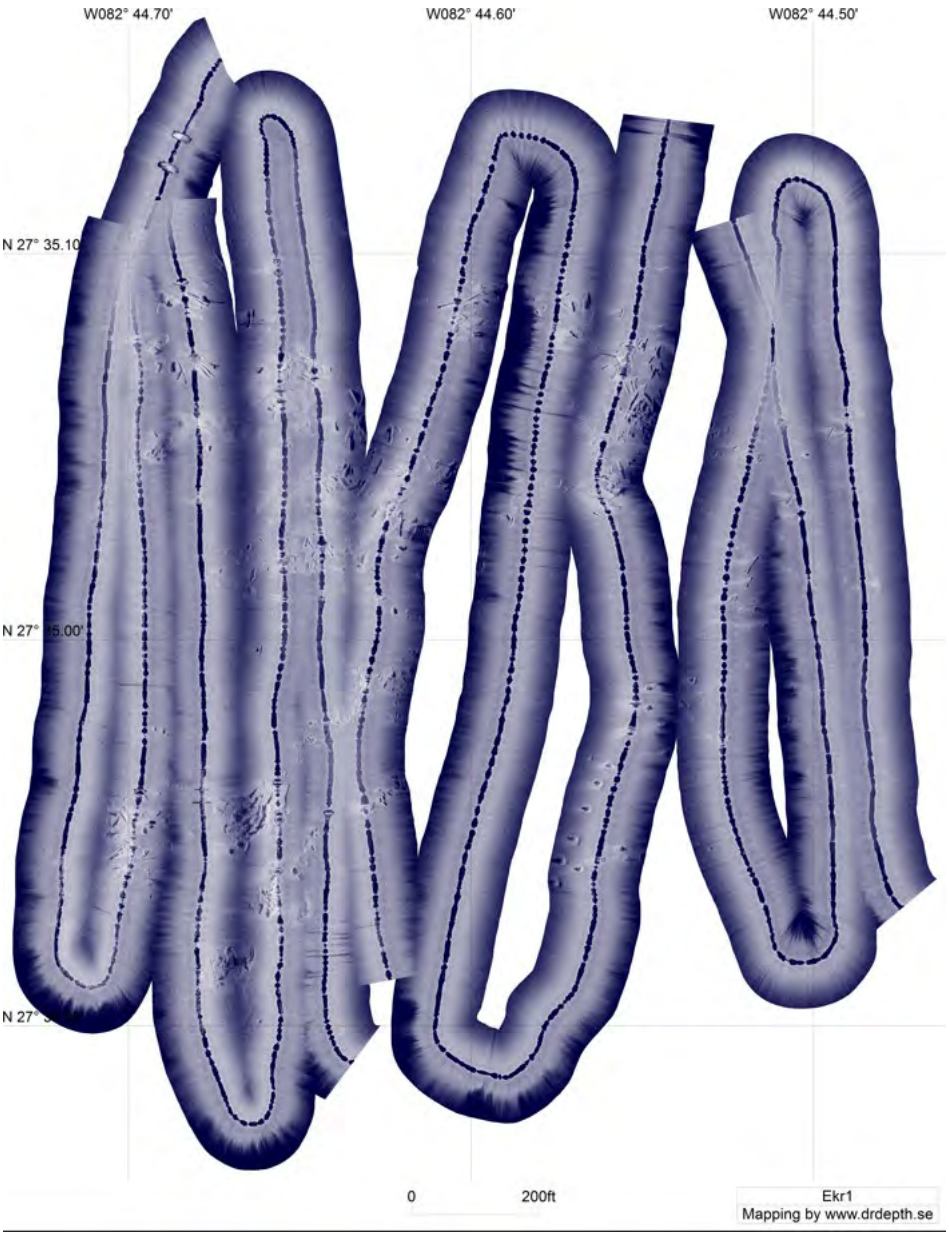


Fig 4. Egmont Key Side scan 05-21-2013 (see below deployments within side scan area).

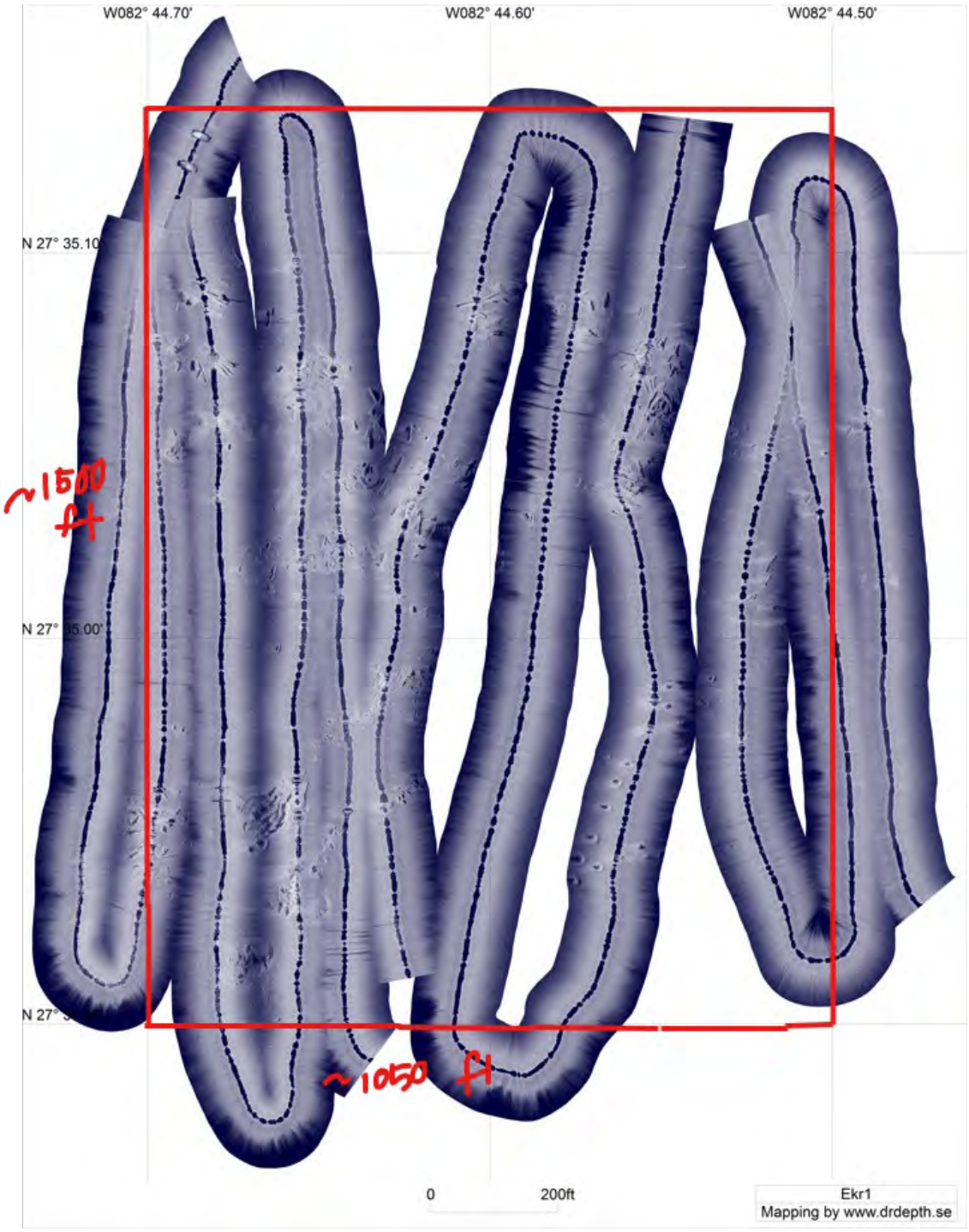


Fig 5. Selected Egmont Key side scan area for estimation from side scan imagery.

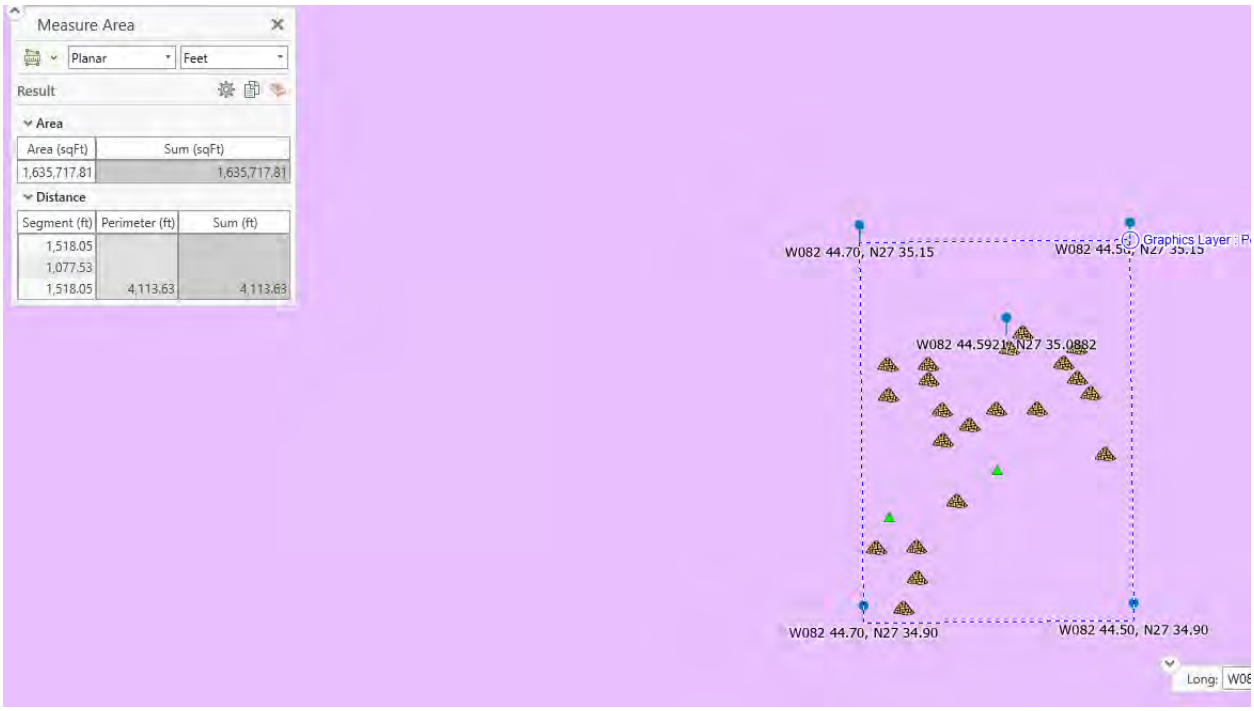


Fig 6. Egmont Key side scan area plotted in GIS with corresponding deployments. Total area was 1,635,718.81 sq ft. Materials estimated to cover about 50% of area mapped.

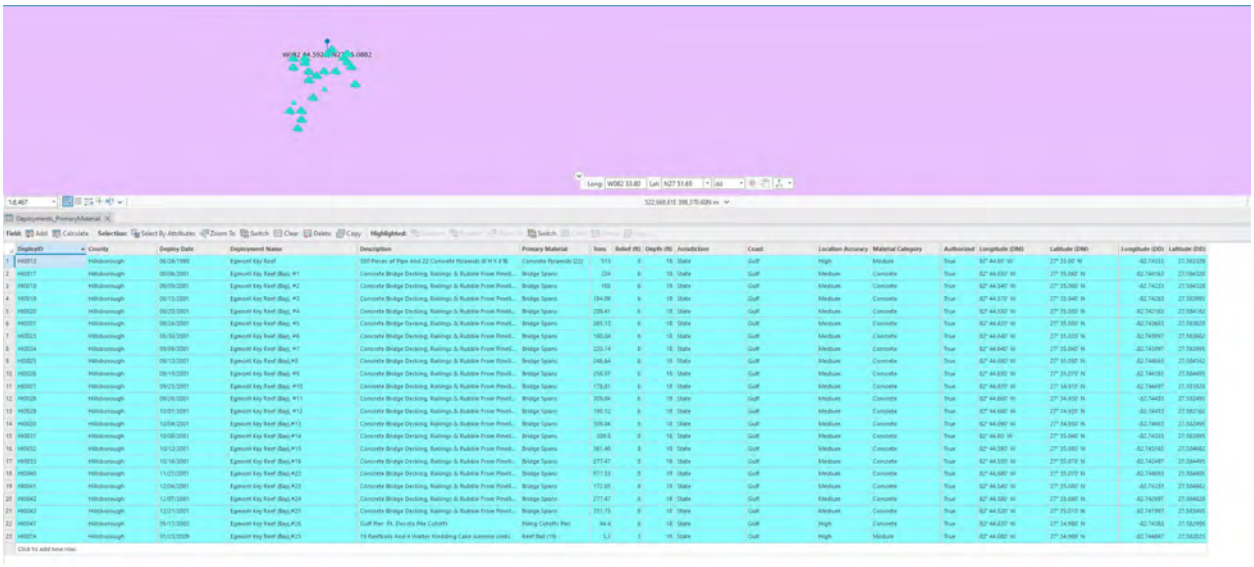


Fig 7. Artificial reefs within Egmont Key side scan; 23 deployments included within area.

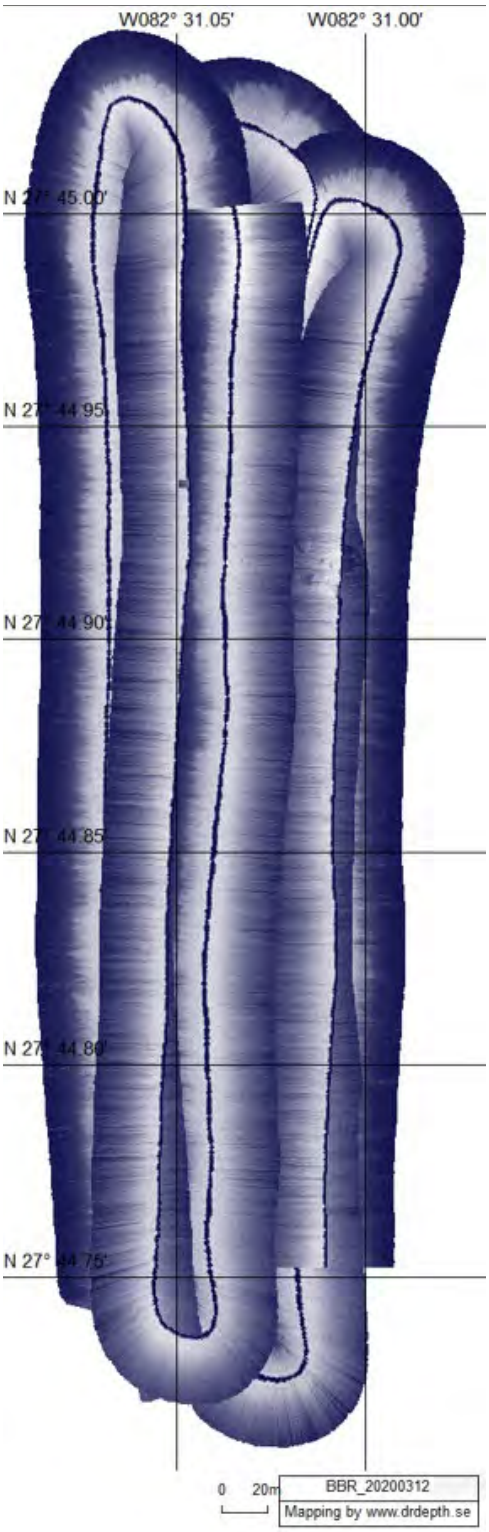


Fig 8. Bahia Bay Reef Side scan 03-12-2020.

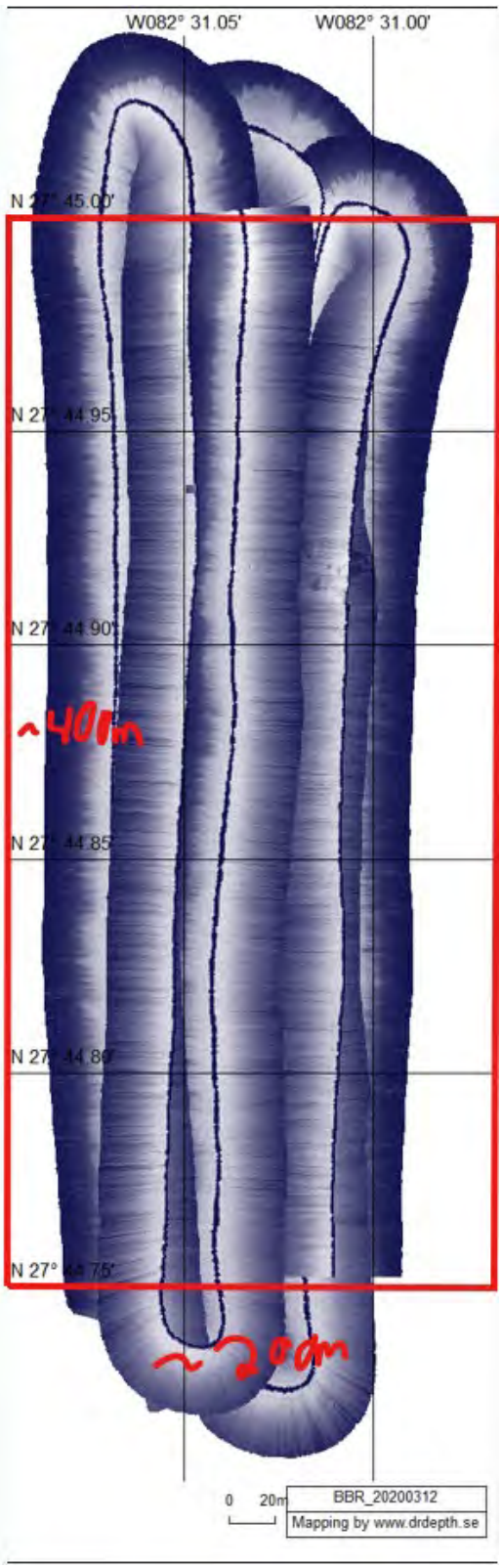


Fig 9. Selected Bahia Bay Reef side scan area for estimation from side scan imagery.

Bahia Reef Deployments

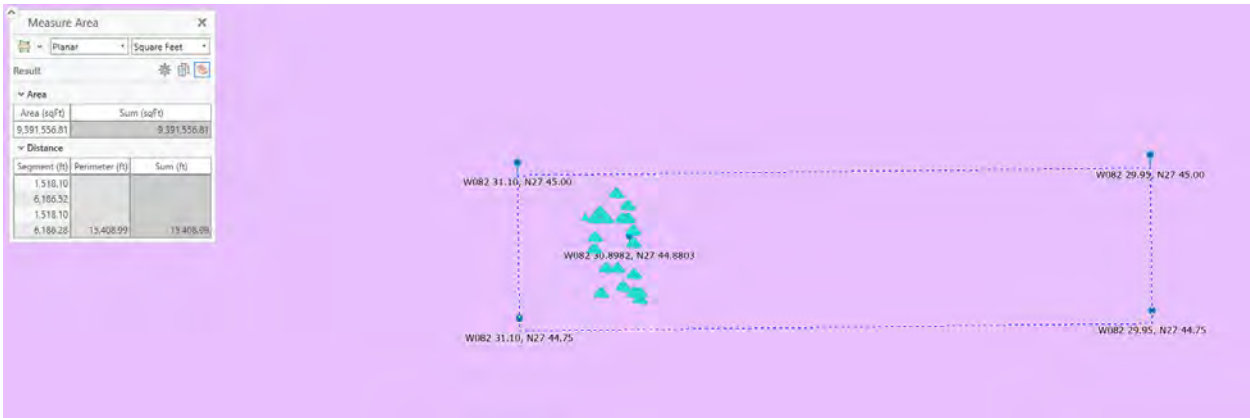


Fig 10. Bahia Bay Reef side scan area plotted in GIS with corresponding deployments. Total area was 9,391,556.81 sq ft. Materials estimated to cover about 1% of area mapped.

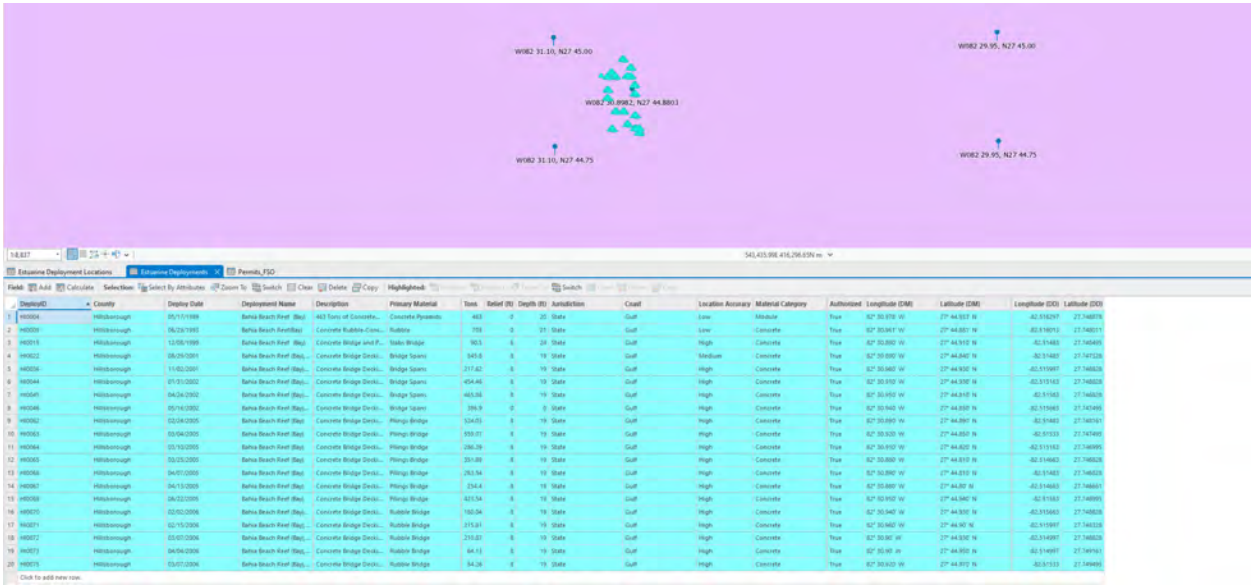


Fig 11. Artificial reefs within Bahia Bay Reef side scan; 20 deployments included in area.

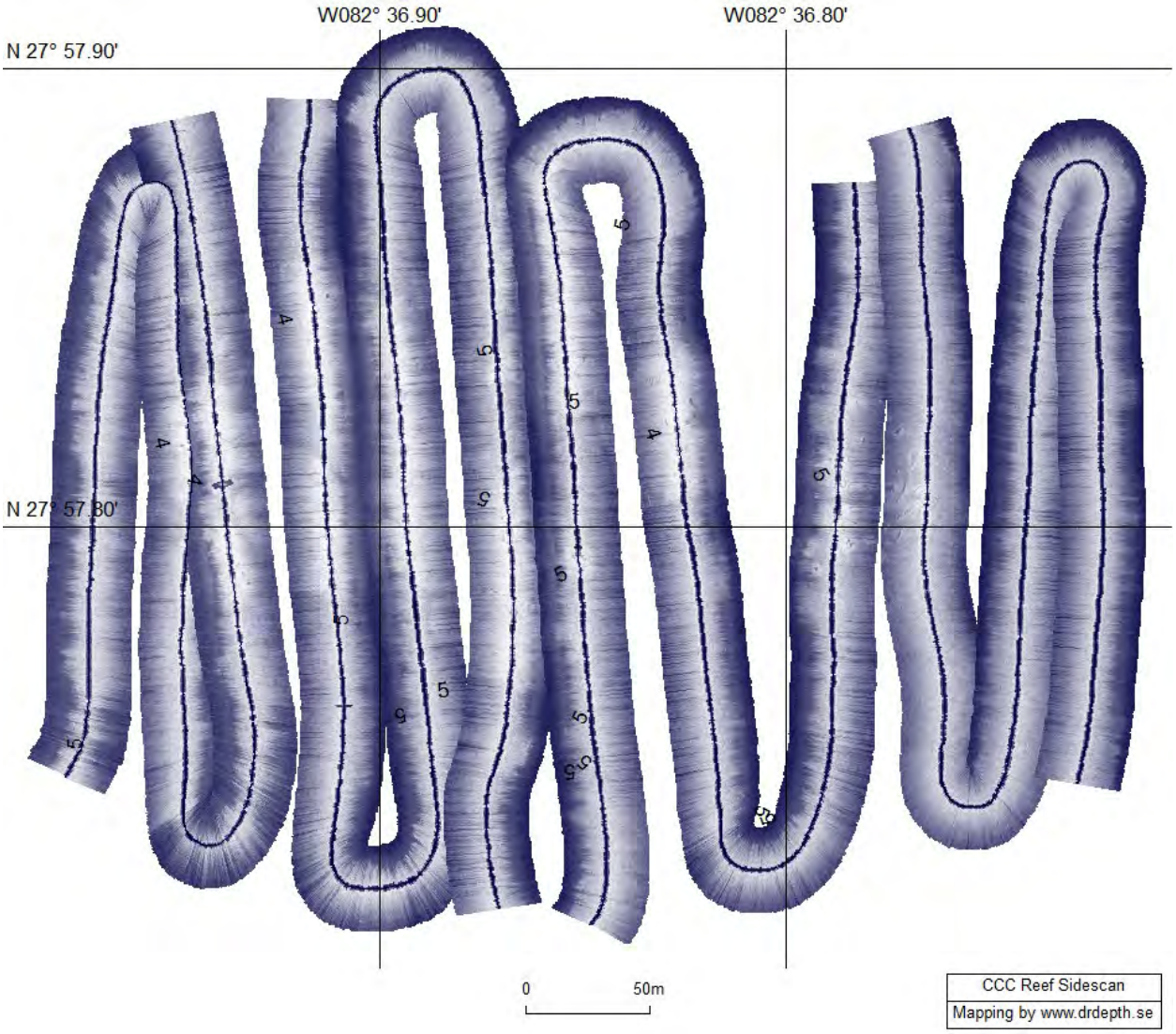


Fig 12. CCR Side scan 01-29-2014, aligns with HI0007 and entire side can extent used in estimate.



Fig 13. CCR side scan area plotted in GIS with corresponding deployment. Total area was 1,956,168.04 sq ft. Materials estimated to cover 1% of area mapped.

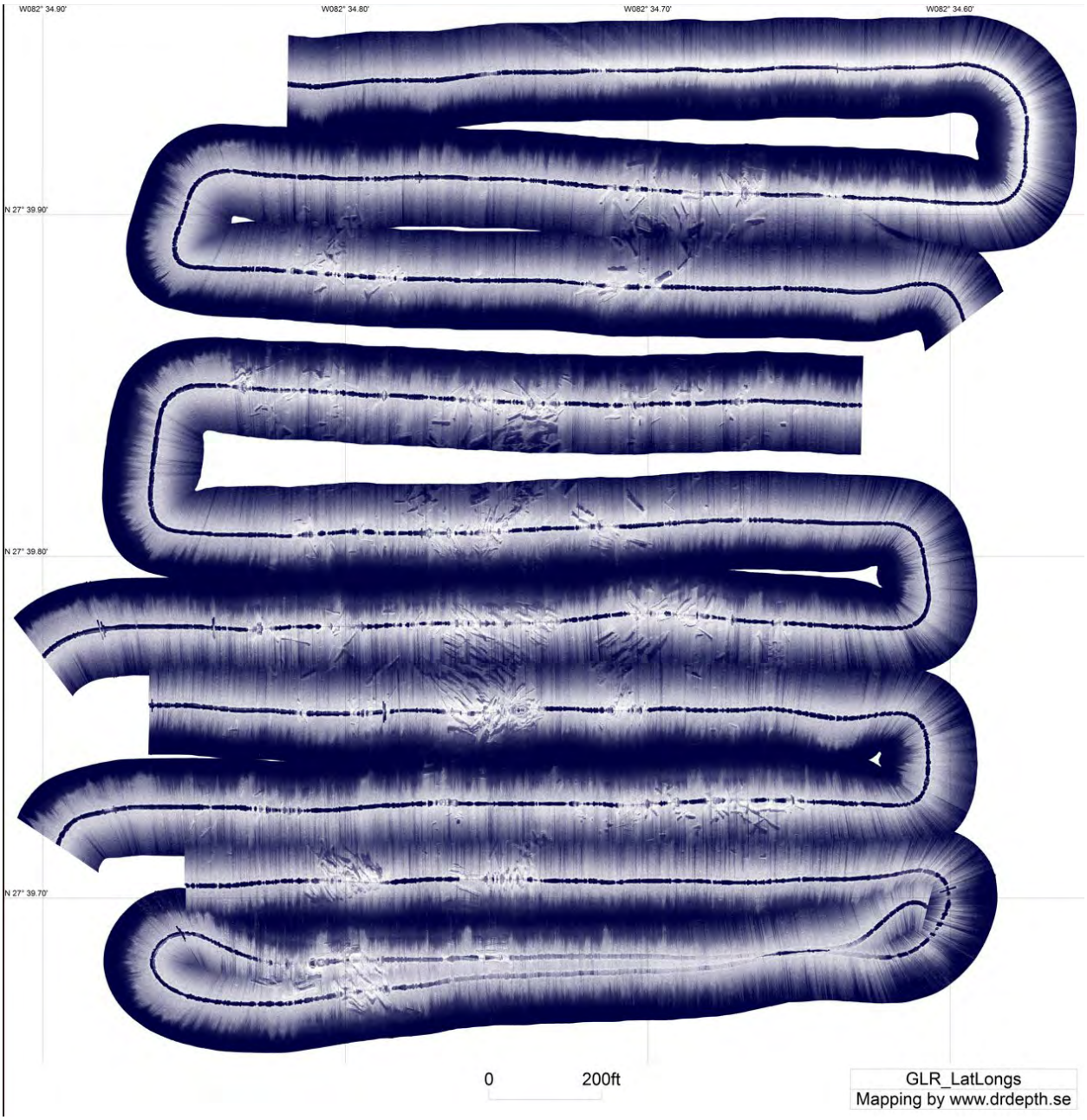


Fig 14. GLR side scan 05-17-2013.

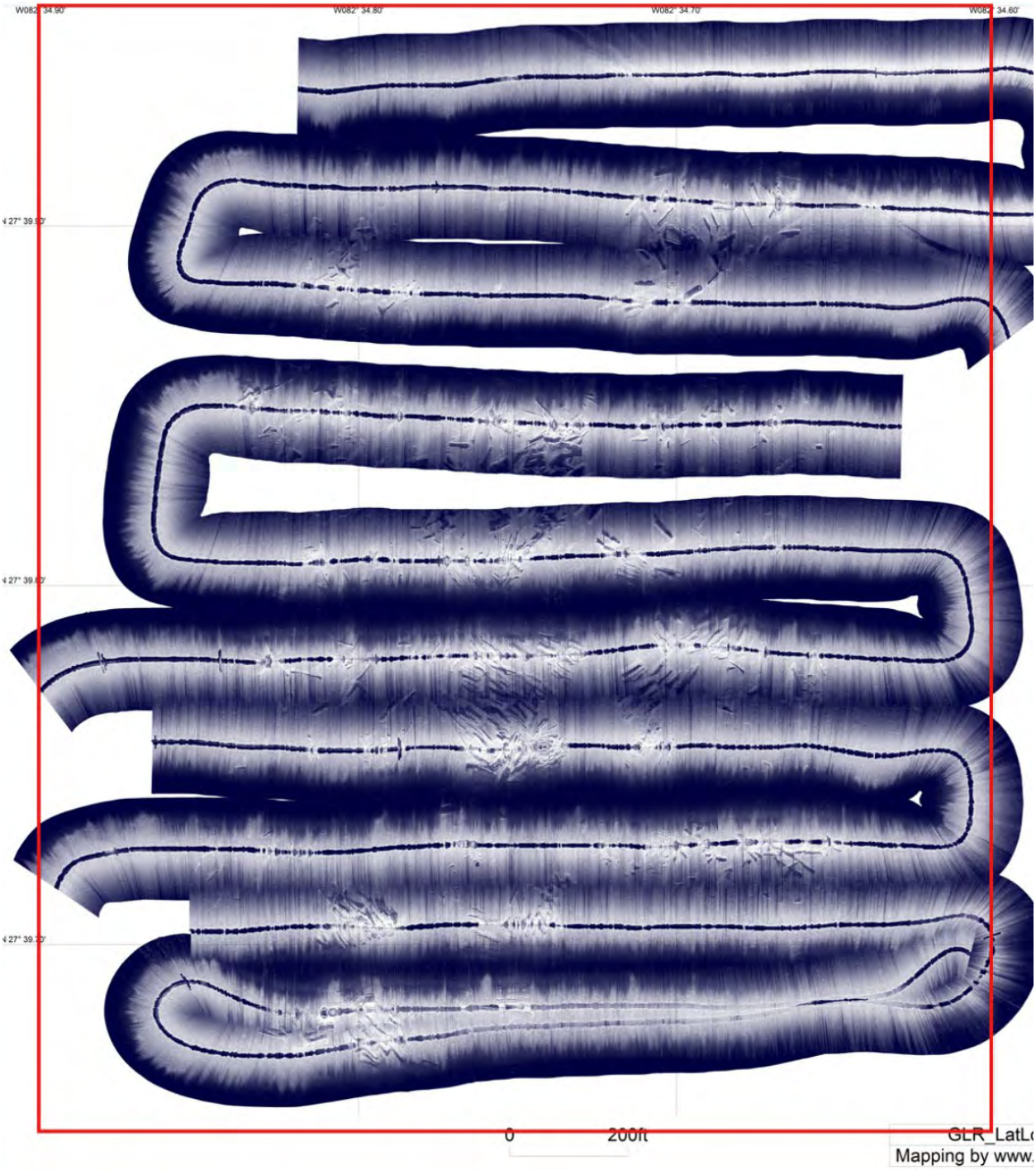


Fig 15. Selected GLR side scan area for estimation from side scan imagery.

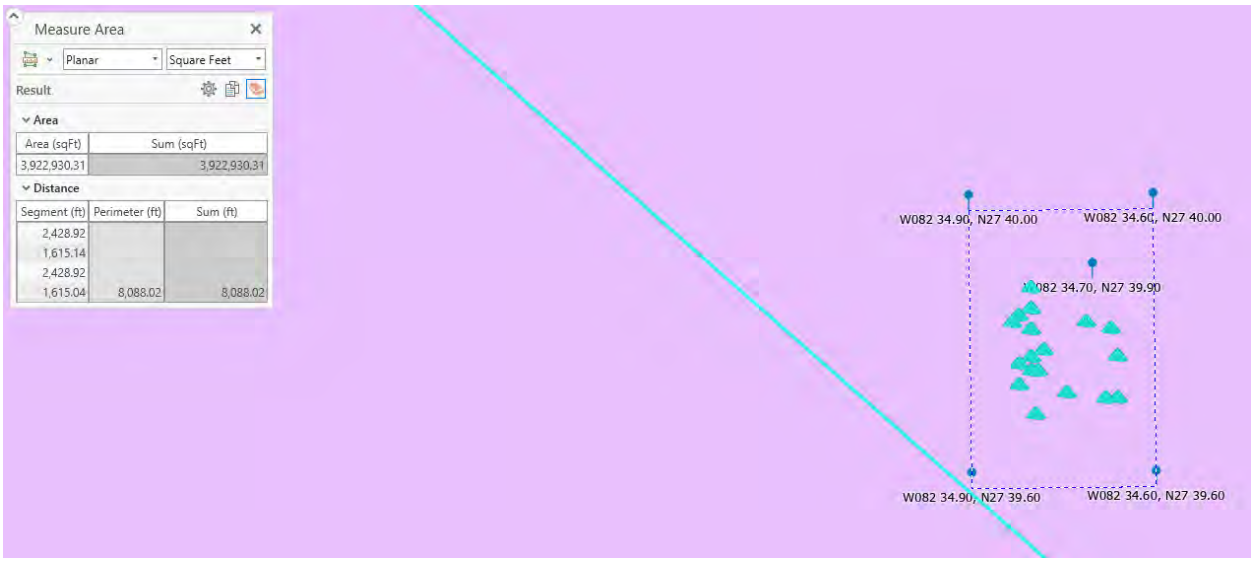


Fig 16. GLR side scan area plotted in GIS with corresponding deployments. Total area was 3,922,930.31 sq ft. Materials estimated to cover about 50% of area mapped.

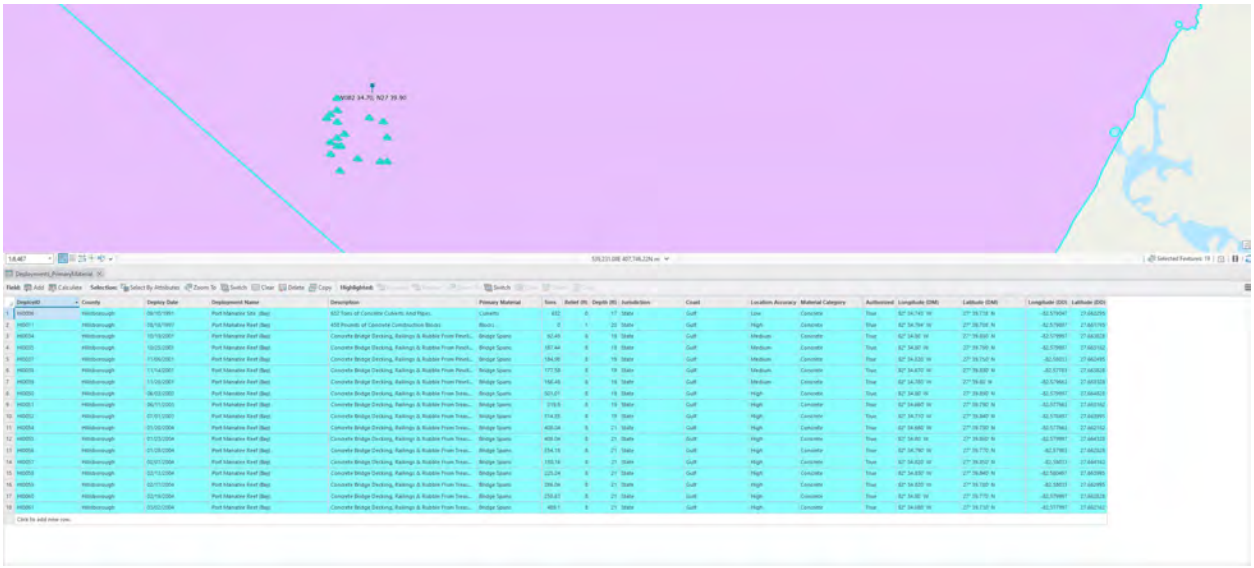


Fig 17. Artificial reefs within GLR side scan; 19 deployments included in area.

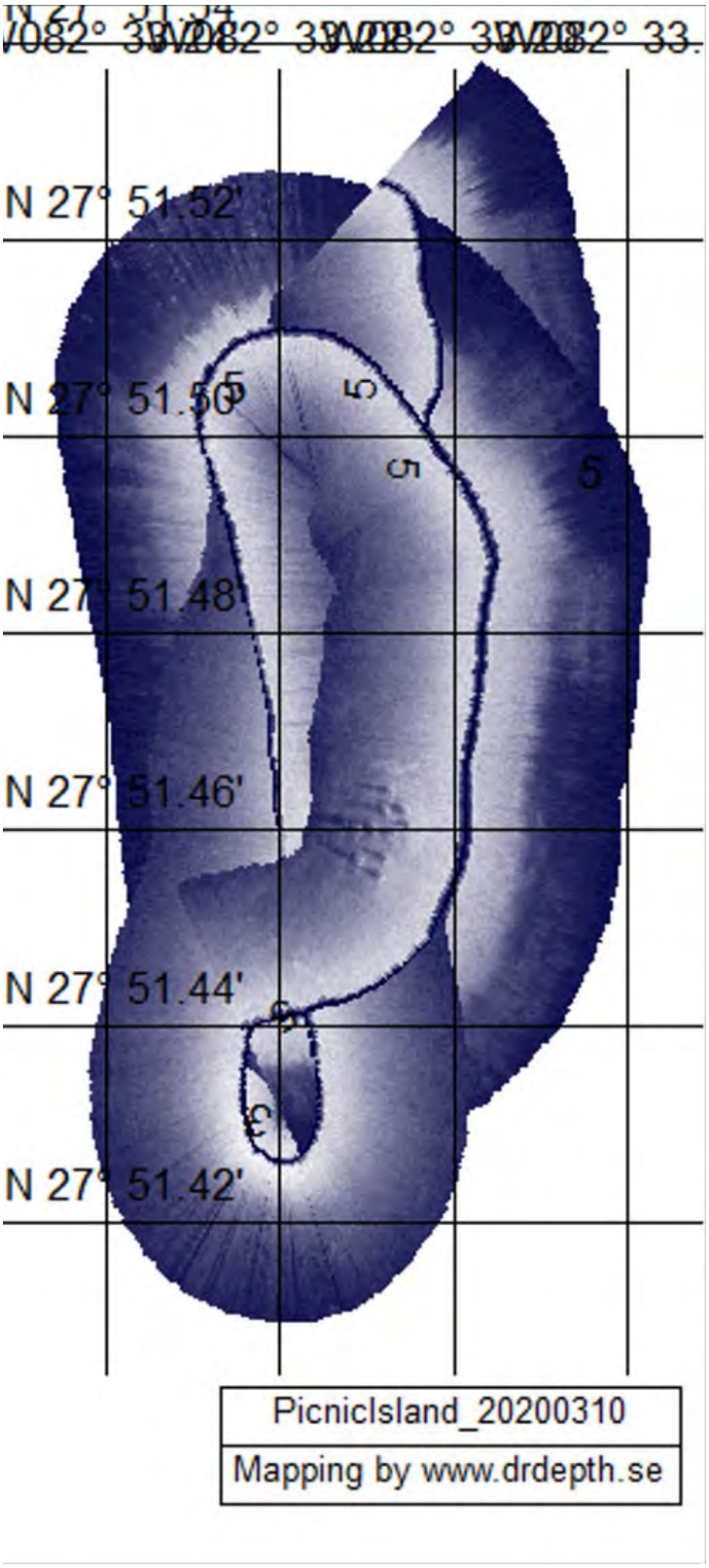


Fig 18. Picnic Island side scan 03-10-2020 for HI0005.

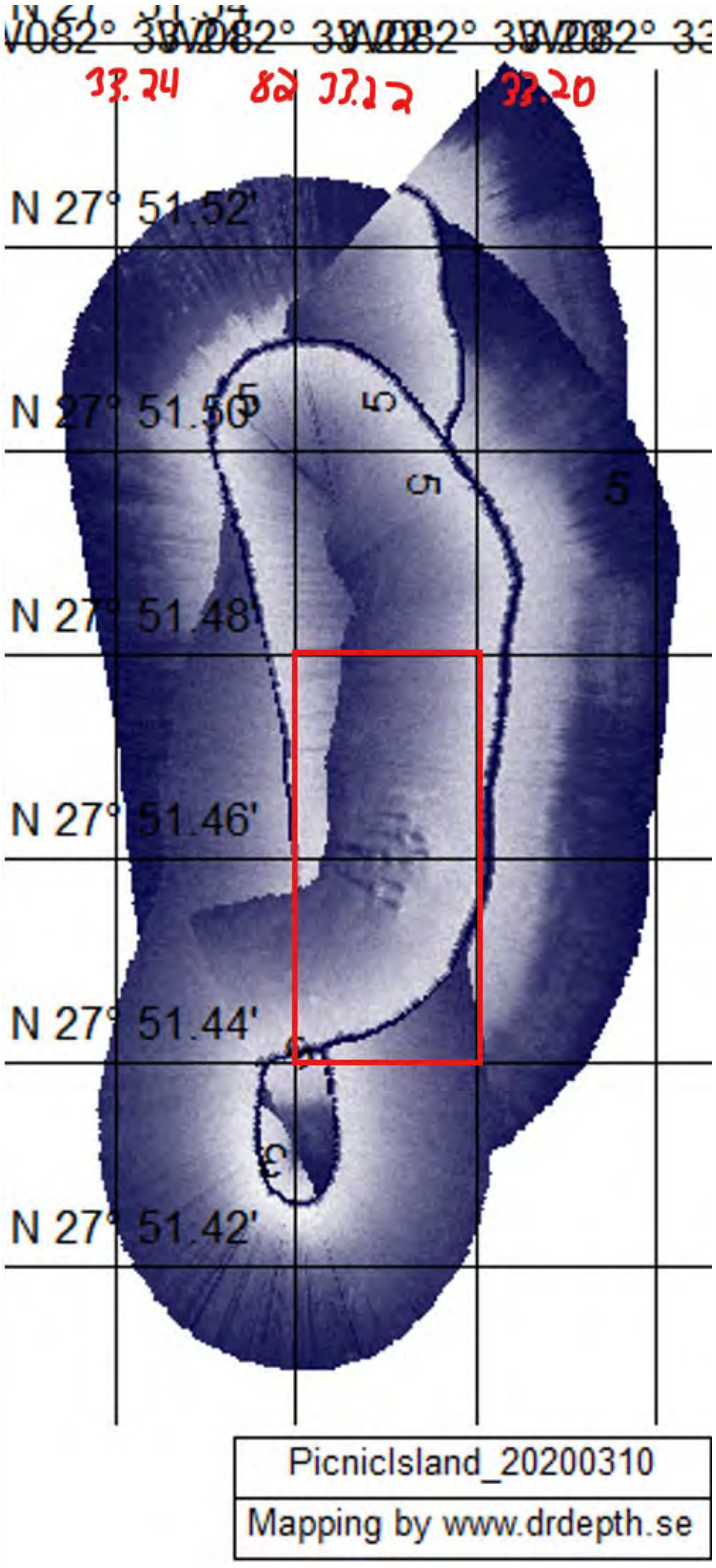


Fig 19. Selected Picnic Island side scan area for estimation from side scan imagery.



Fig 20. Picnic Island side scan area plotted in GIS with corresponding deployments. Total area was 26,107.20 sq ft. Materials estimated to cover about 25% of area mapped.

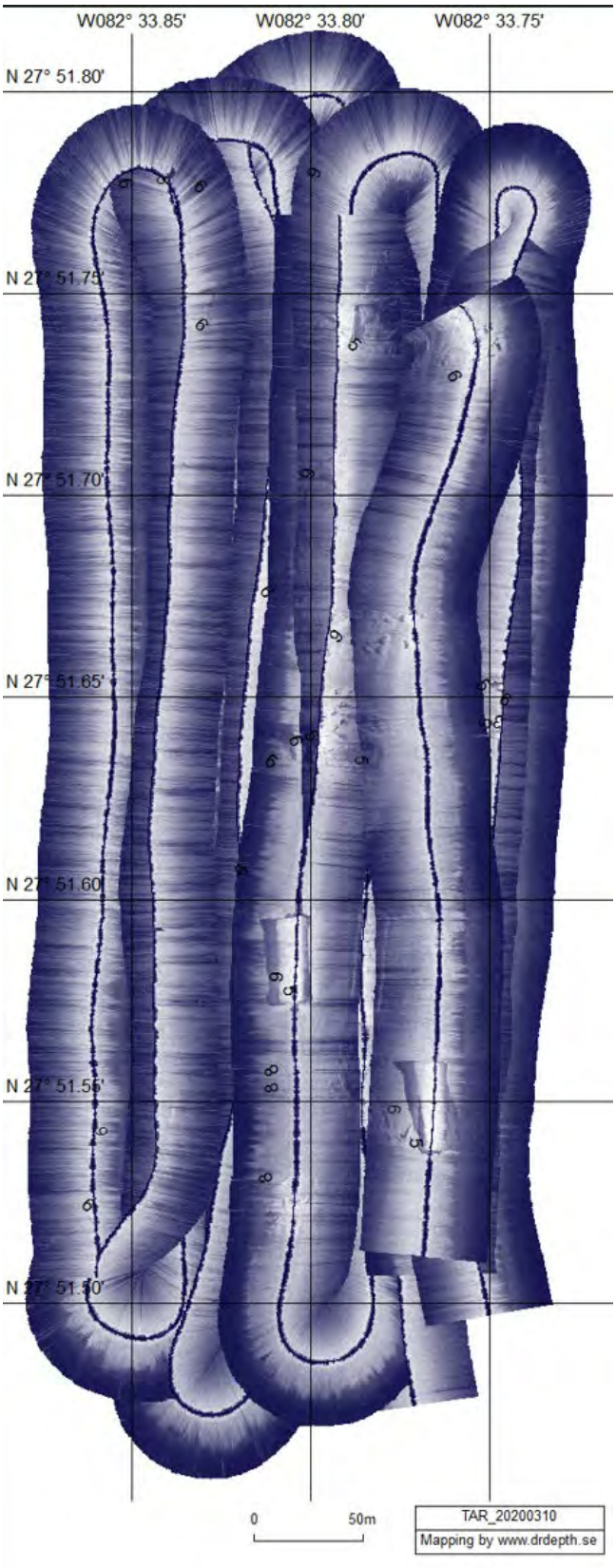


Fig 21. Ted Adams Reef side scan 03-10-2020.

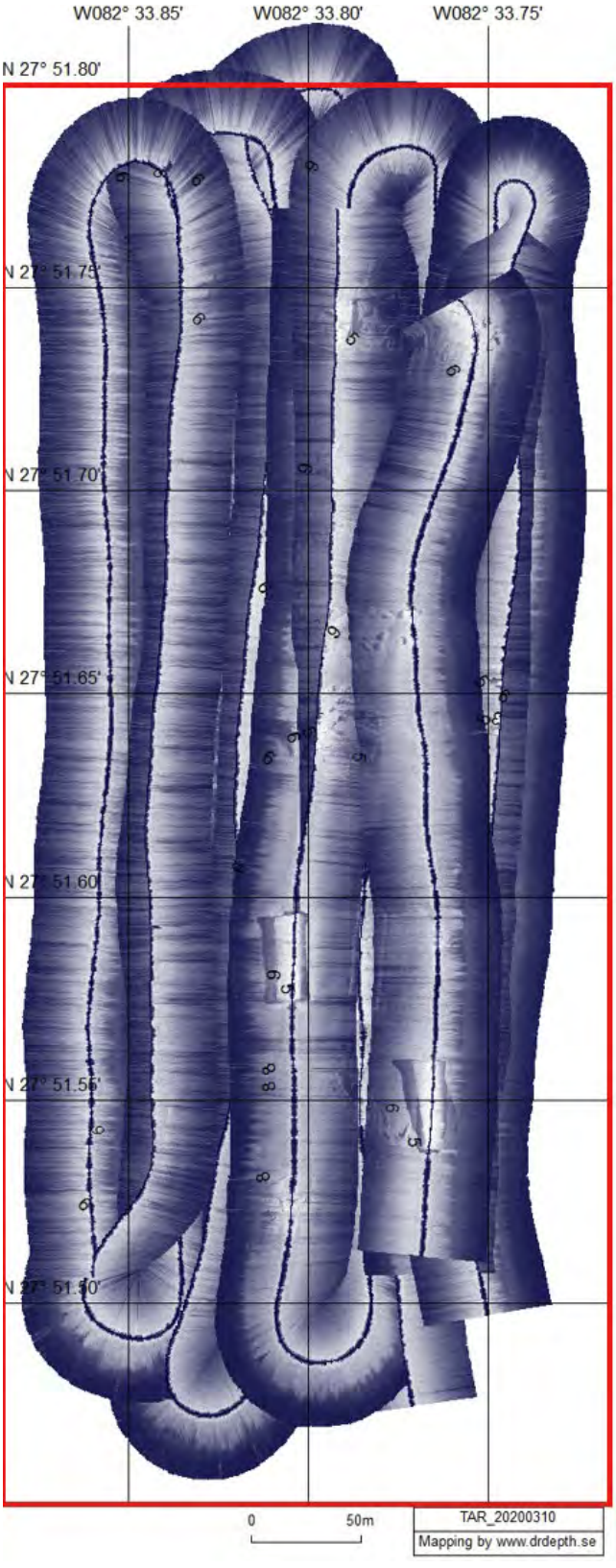


Fig 22. Selected Ted Adams Reef side scan area for estimation from side scan imagery.



Fig 23. Ted Adams Reef side scan area plotted in GIS with corresponding deployments. Total area was 2,284,323.16 sq ft. Materials are estimated to cover about 5% of area mapped.



Fig 24. Artificial reefs within Ted Adams Reef side scan; 8 deployments included in area.

Manatee County Deployments:

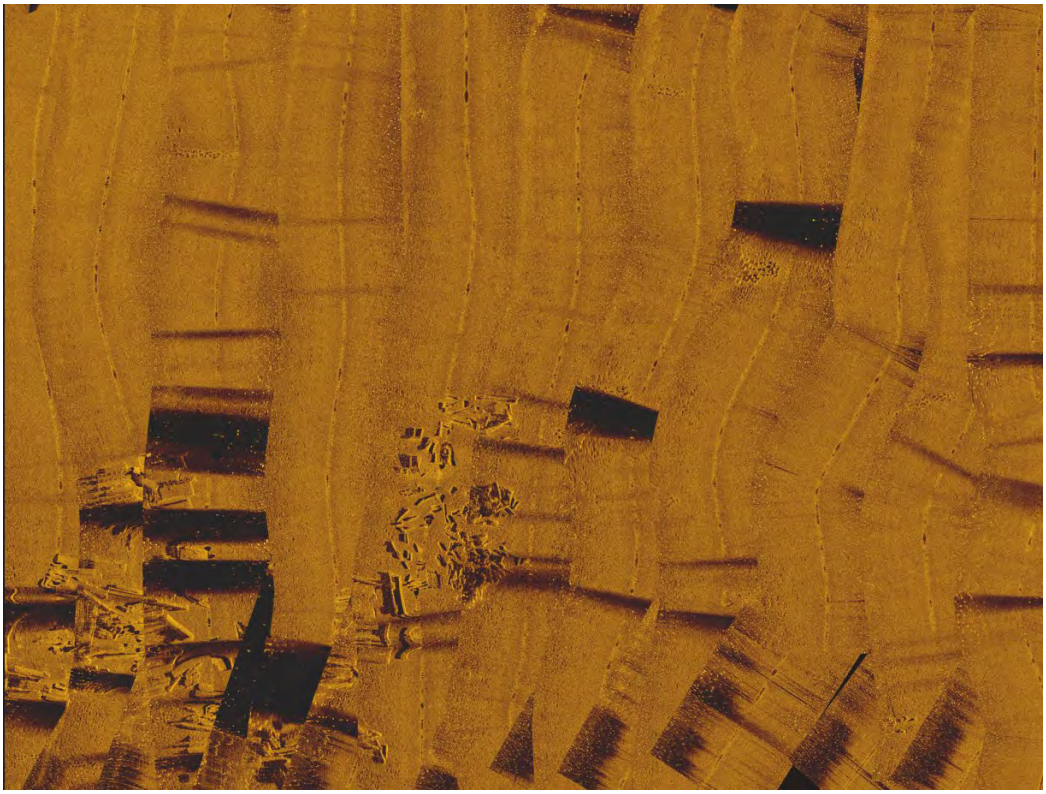


Fig 25. Manatee County side scan used for estimate. Side scan takes up entire dimensions of Southeast Tampa Bay Bridge Reef permitted area. Materials are estimated as 25% of area.

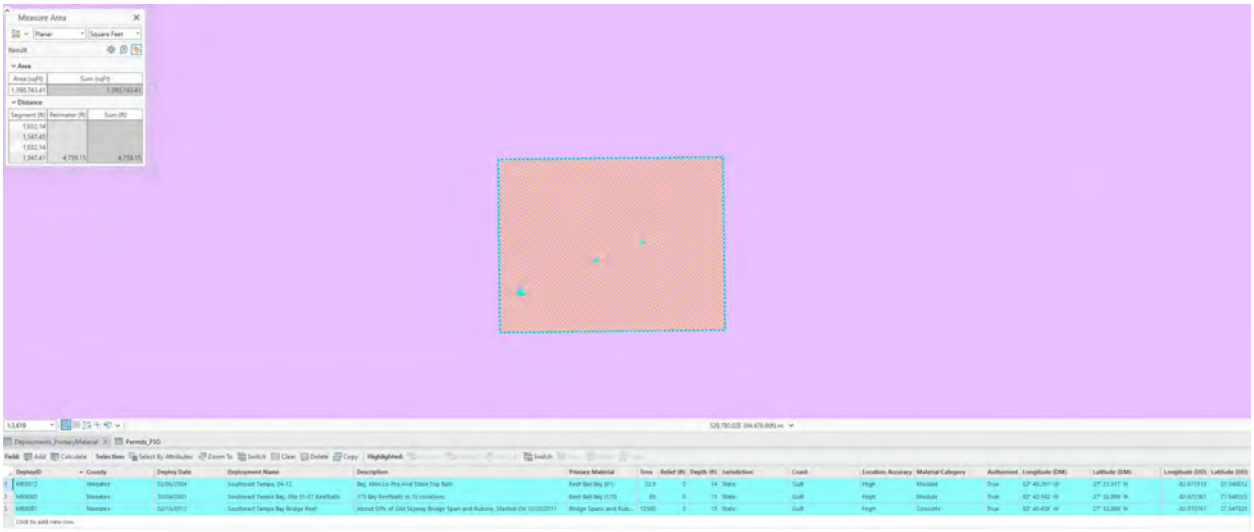


Fig 26. Manatee County side scan area plotted in GIS with corresponding deployments. Total area was 1,390,743.41 sq ft. Materials estimated to cover about 50% of area mapped. There are 32 deployments within permitted area.

Miami-Dade County Deployments:

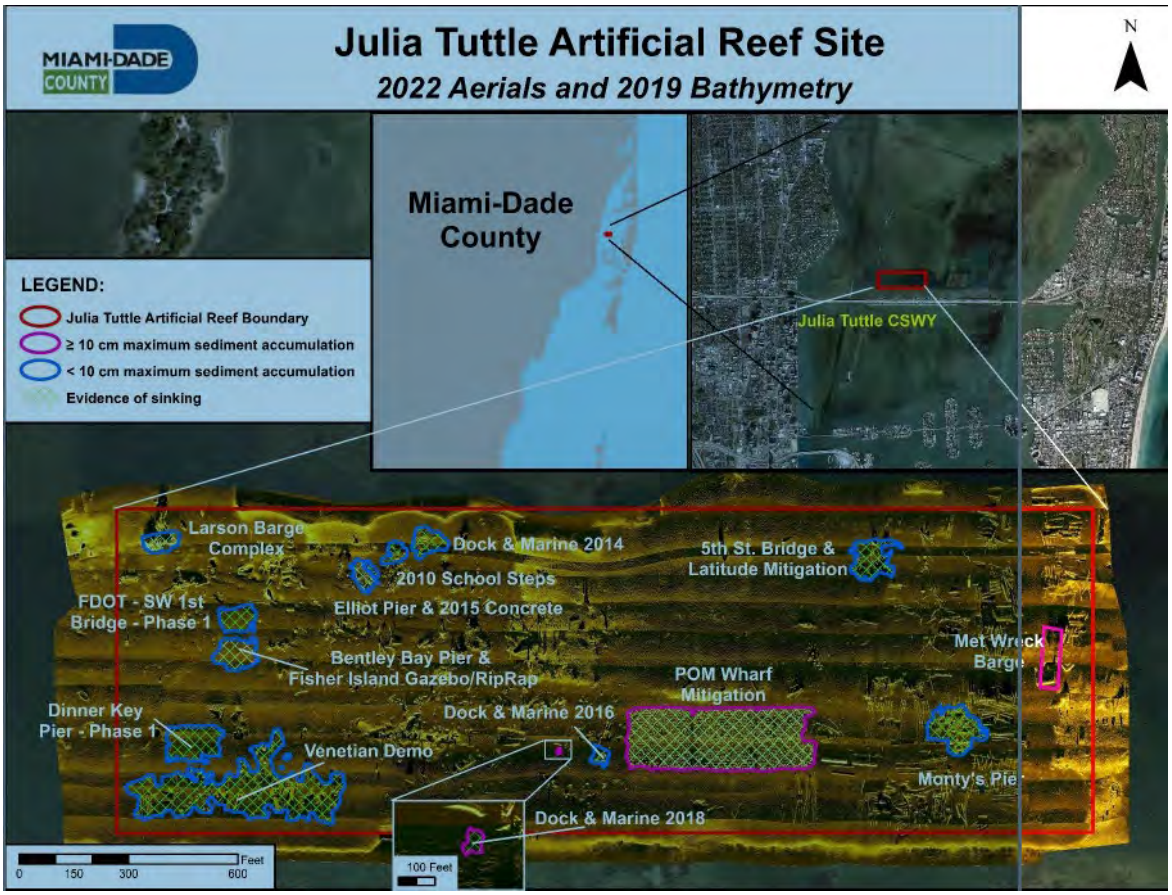


Fig 27. Julia Tuttle Bathymetry from 2025 Miami-Dade Report.



Fig 28. Julia Tuttle permitted area was 2,381,893.09 sq ft. Materials estimated to cover about 25% of area mapped. ME0087, ME0082, and ME0072 within permitted area.



Fig 29. Brickell Inshore Reef Bathymetry from 2025 Miami-Dade Report.



Fig 30. Brickell Inshore Reef permitted area was 444,654.91 sq ft. Materials estimated to cover about 75% of area mapped. There are 4 deployments within permitted area.

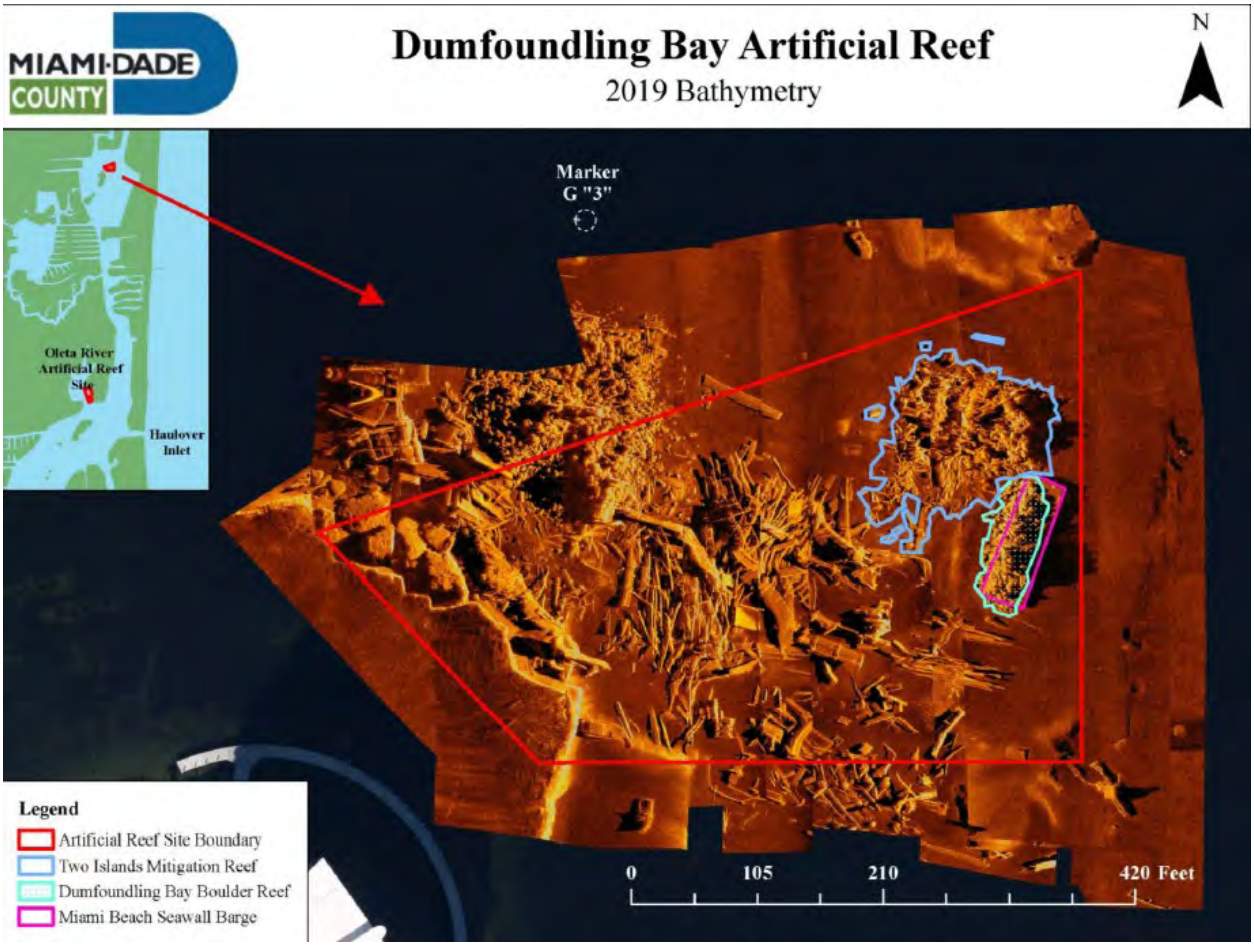


Fig 31. Dumfounding Bay Bathymetry from 2025 Miami-Dade Report.

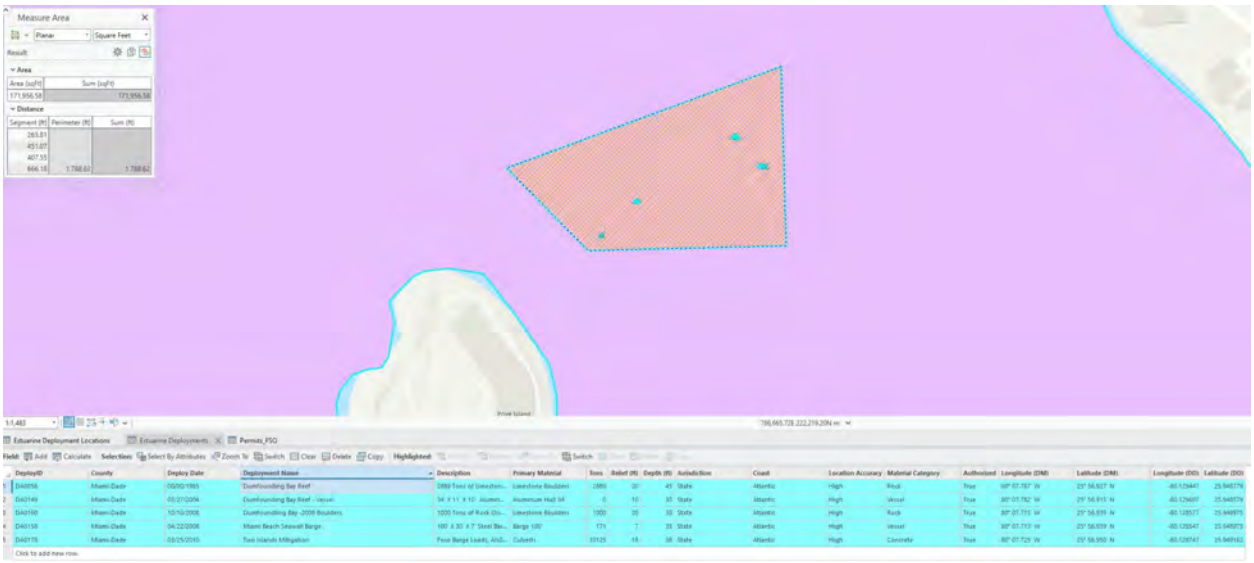


Fig 32. Dumfounding Bay permitted area was 171,956.58 sq ft. Materials estimated to cover full permit area. There are 5 deployments within permitted area.

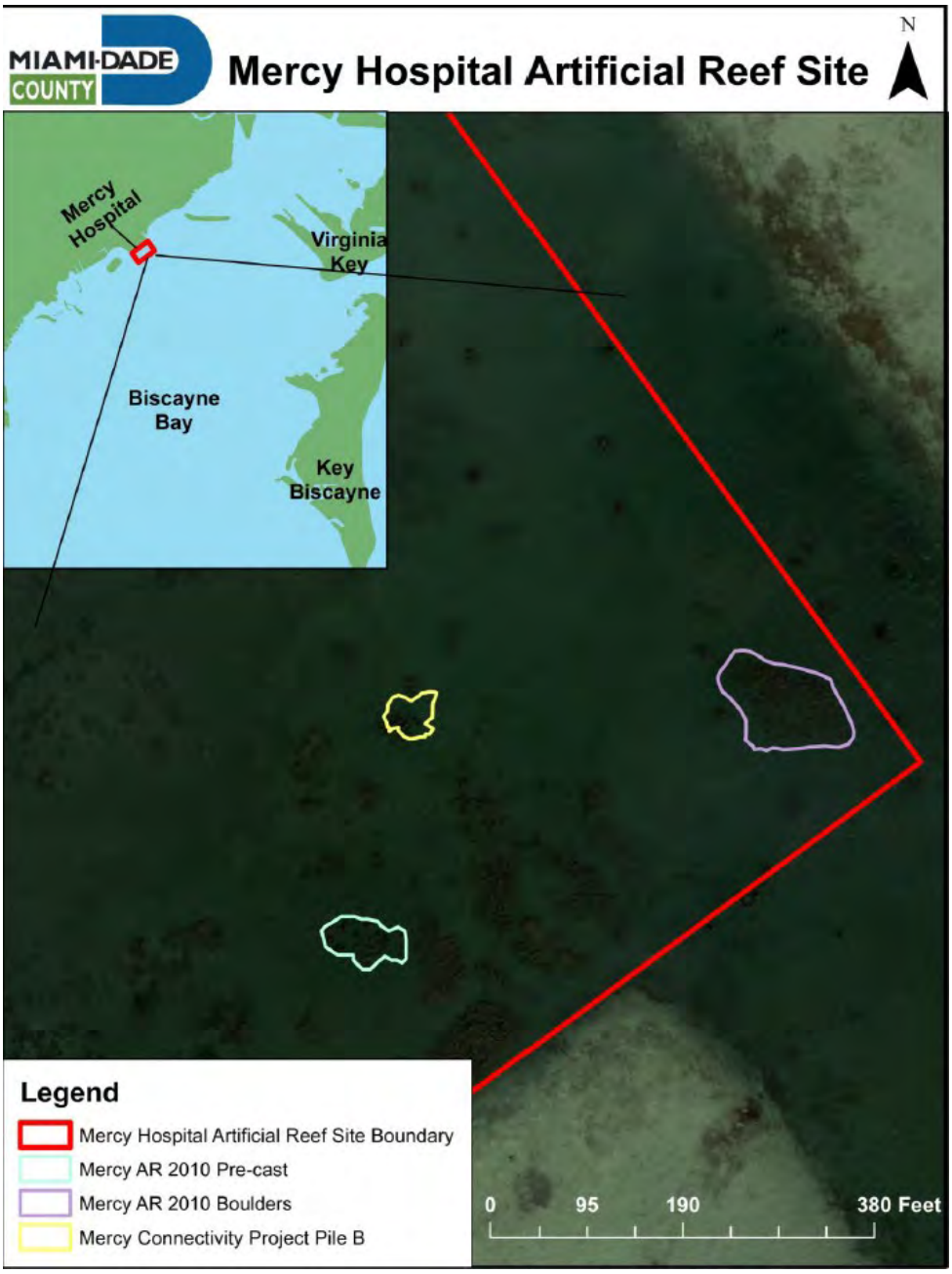


Fig 33. Mercy Hospital Bathymetry from 2025 Miami-Dade Report.

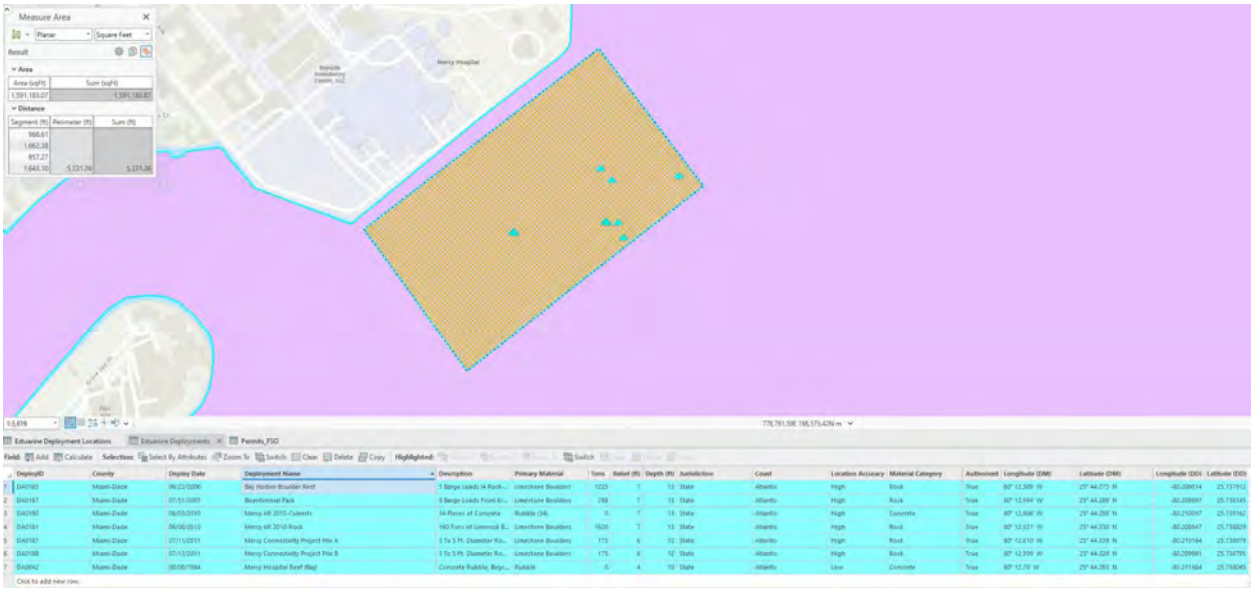


Fig 34. Mercy Hospital permitted area was 1,591,183.07 sq ft. Materials estimated to cover 10% of permit area. There are 5 deployments within permitted area.

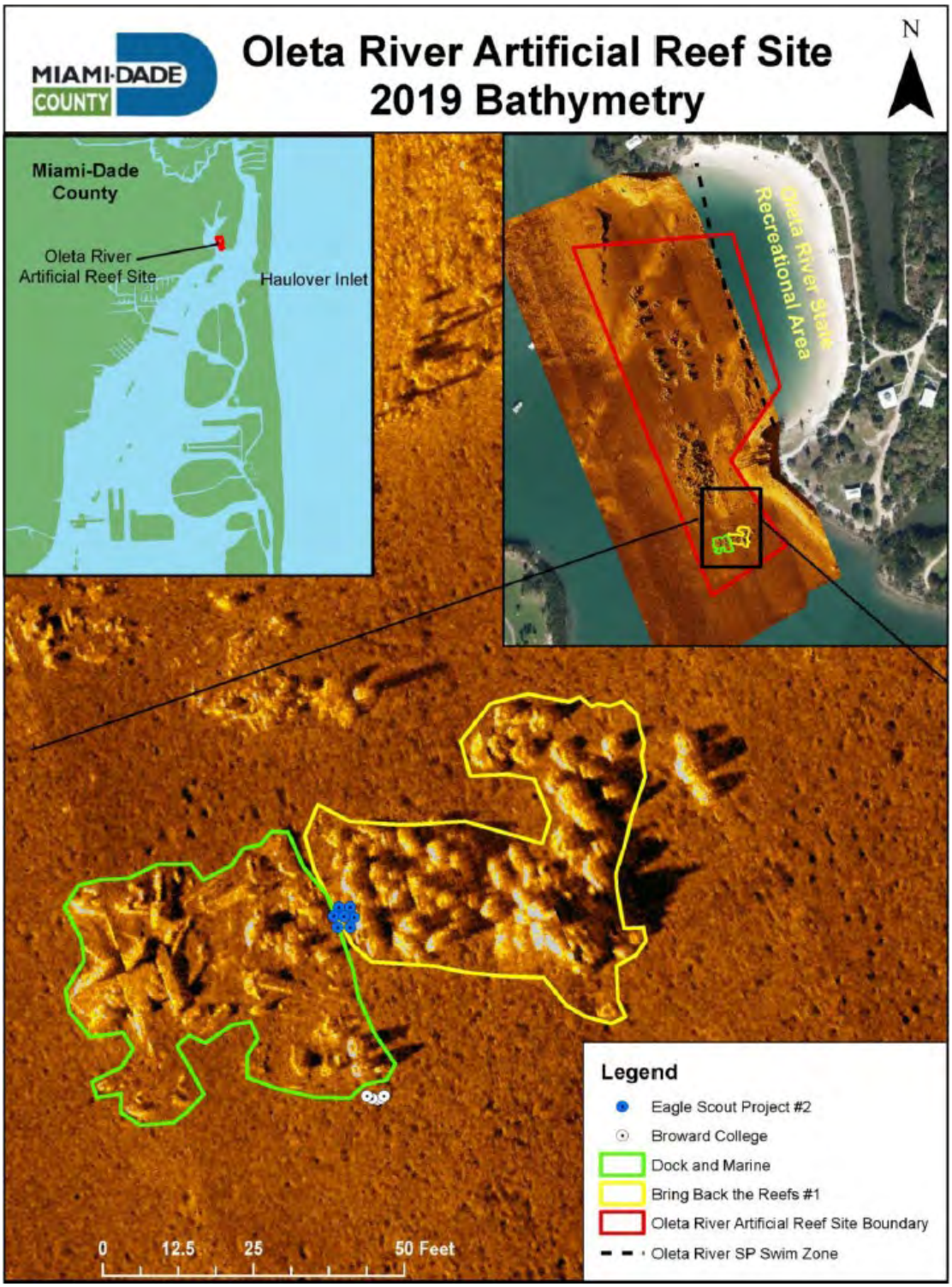


Fig 35. Oleta River Bathymetry from 2025 Miami-Dade Report.

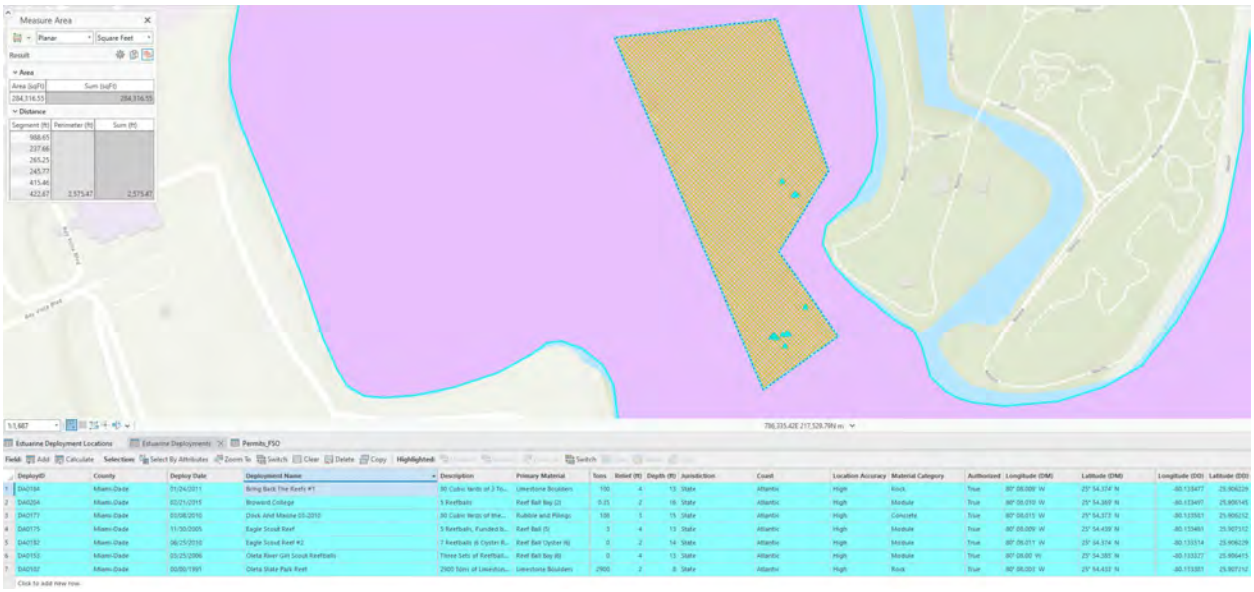


Fig 36. Oleta River permitted area was 284,316.55 sq ft. Materials estimated to cover 10% of permit area. There are 7 deployments within permitted area.

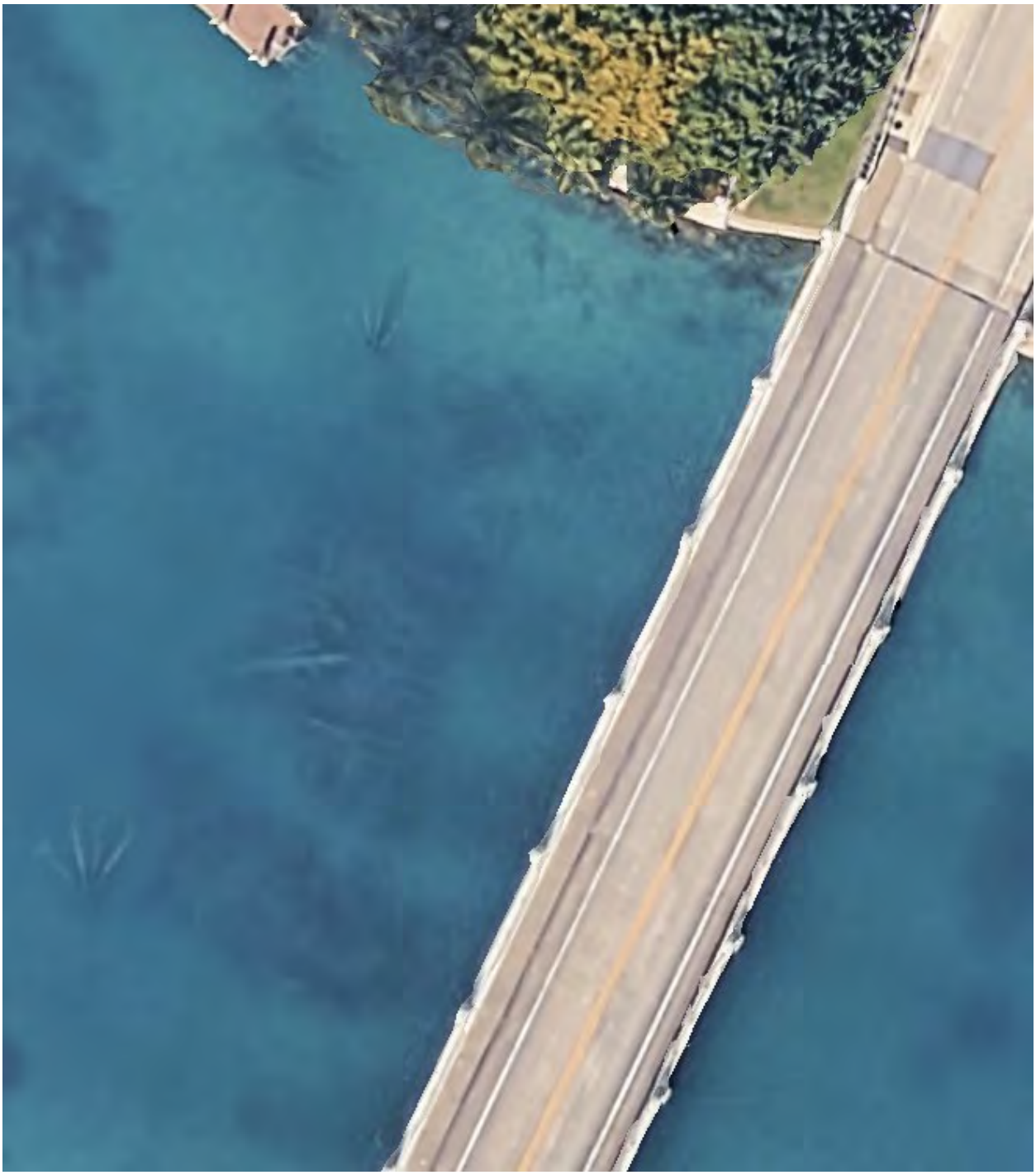


Fig 37. Aerial of artificial reef deployment off Star Island.

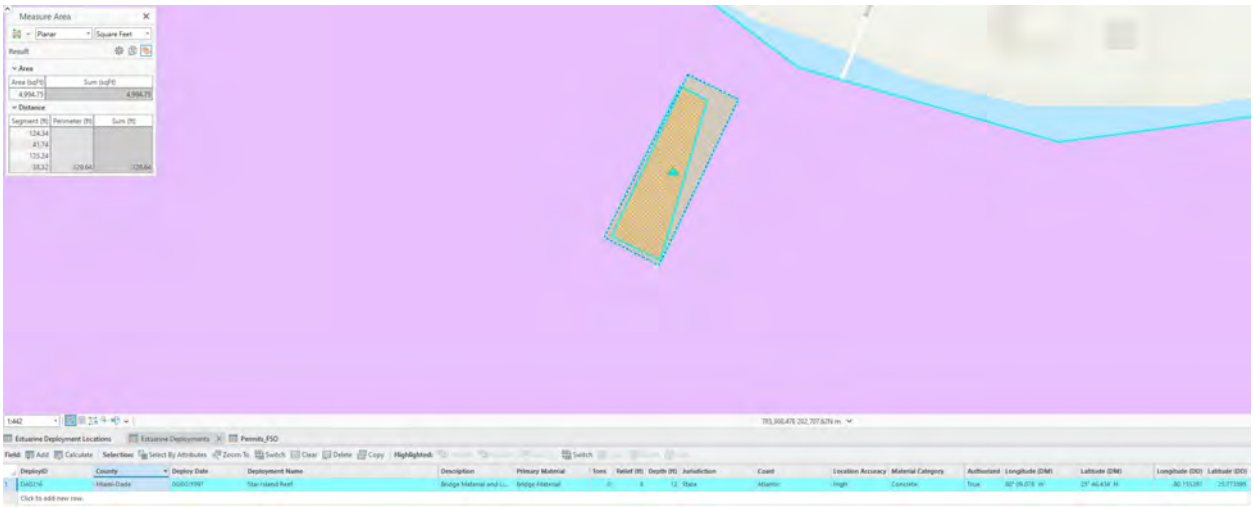


Fig 38. Star Island permitted area was 4,994.75 sq ft. Materials estimated as 90% of permit area.

Palm Beach County Deployments:



Fig 39. Sugar Sands permitted area, encompassing 2,443,383.37 sq ft. Deployments estimated to encompass 5%.

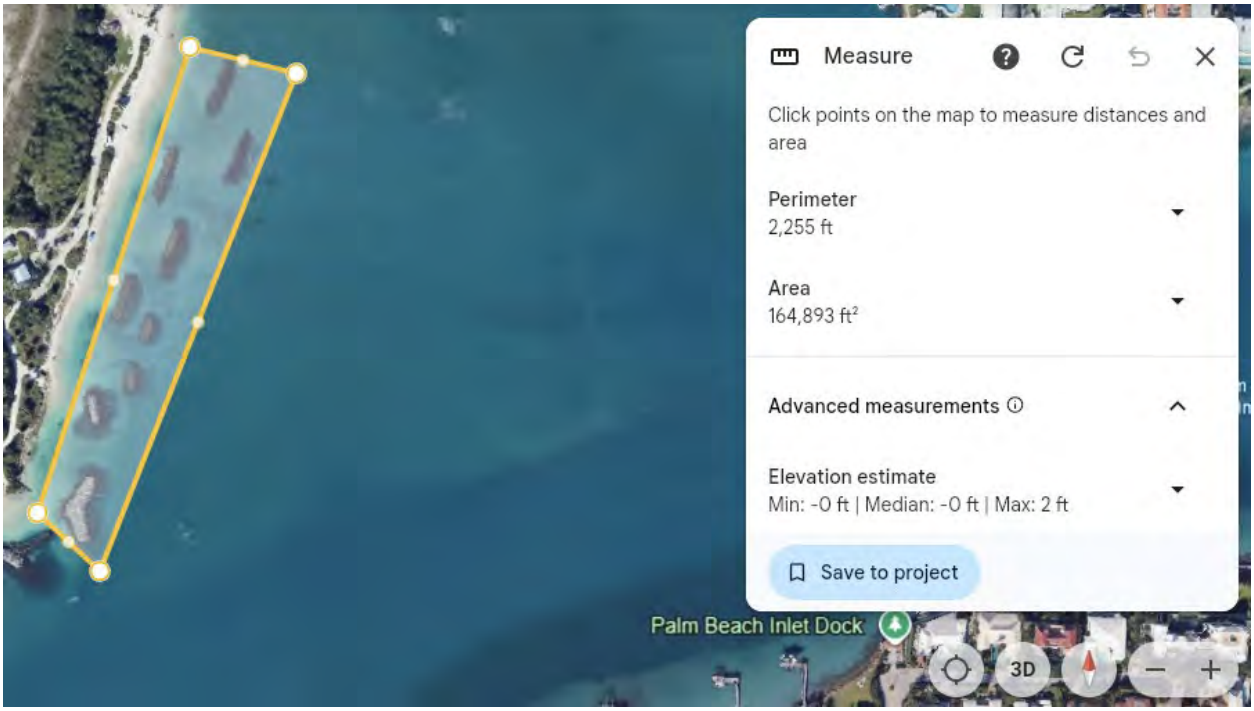


Fig 40. Peanut Island deployment area from aerial, encompassing 164,893 sq ft. Estimated as 25% of area.

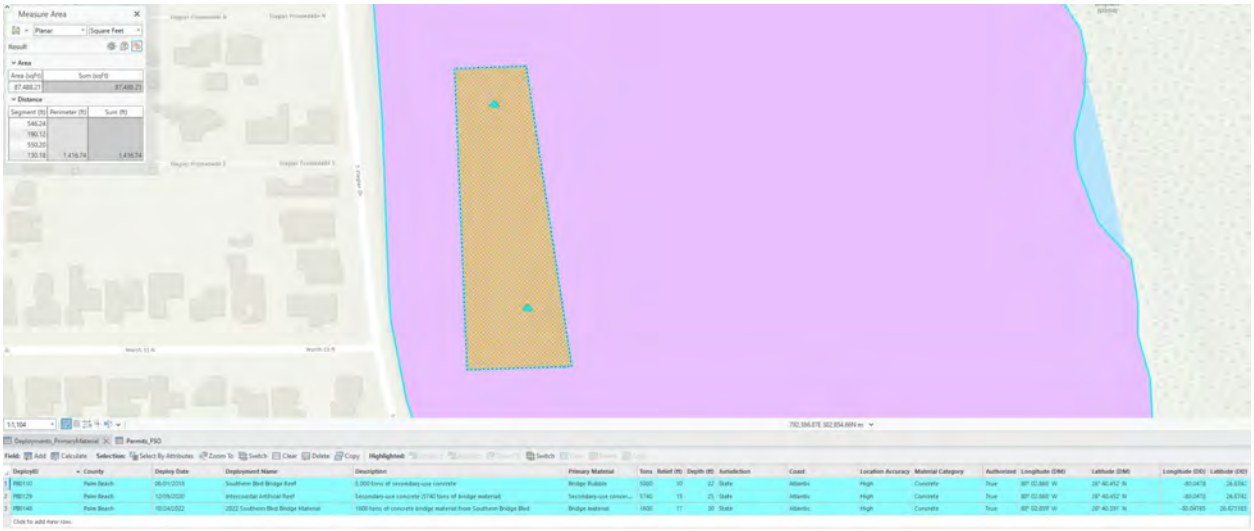


Fig 41. Southern Blvd permit area encompassing 87,488.21 sq ft. Deployments estimated to encompass permitted area since each have high tonnage.

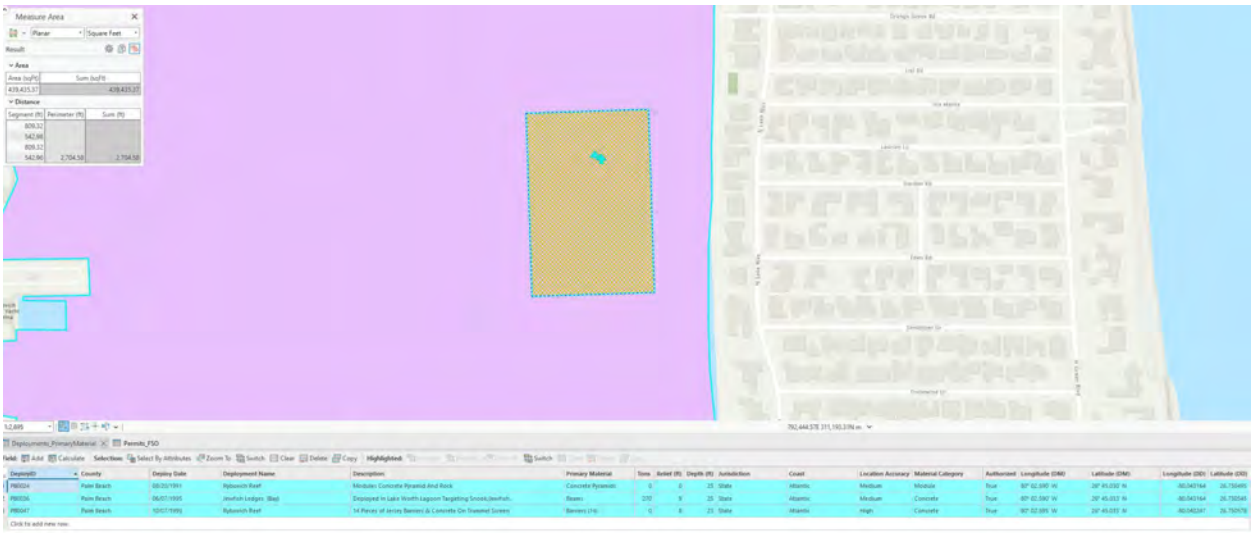


Fig 42. Rybovich Reef permitted area, encompassing 439,435.37 sq ft. Deployments estimated as 5% of permitted area.

Pinellas County Deployments:

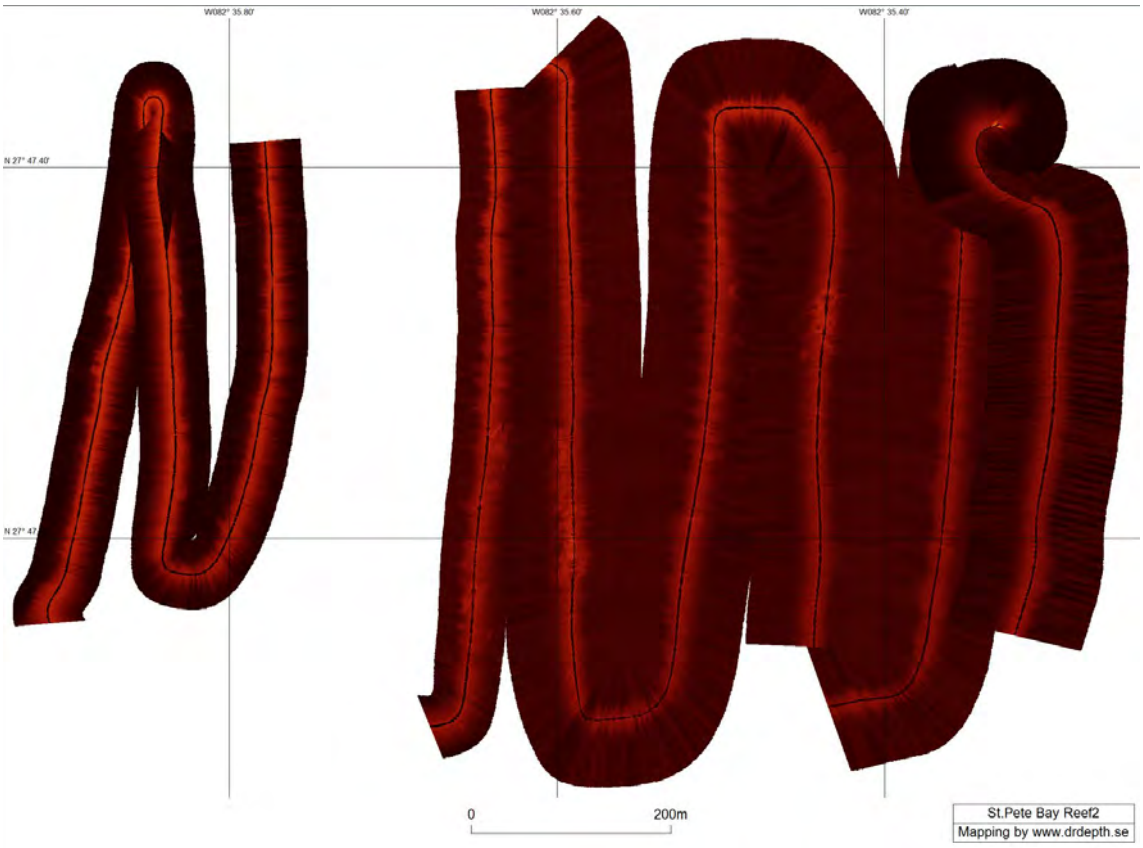


Fig 43. St Pete Bay Reef (Side scan date unknown), for PI0107.

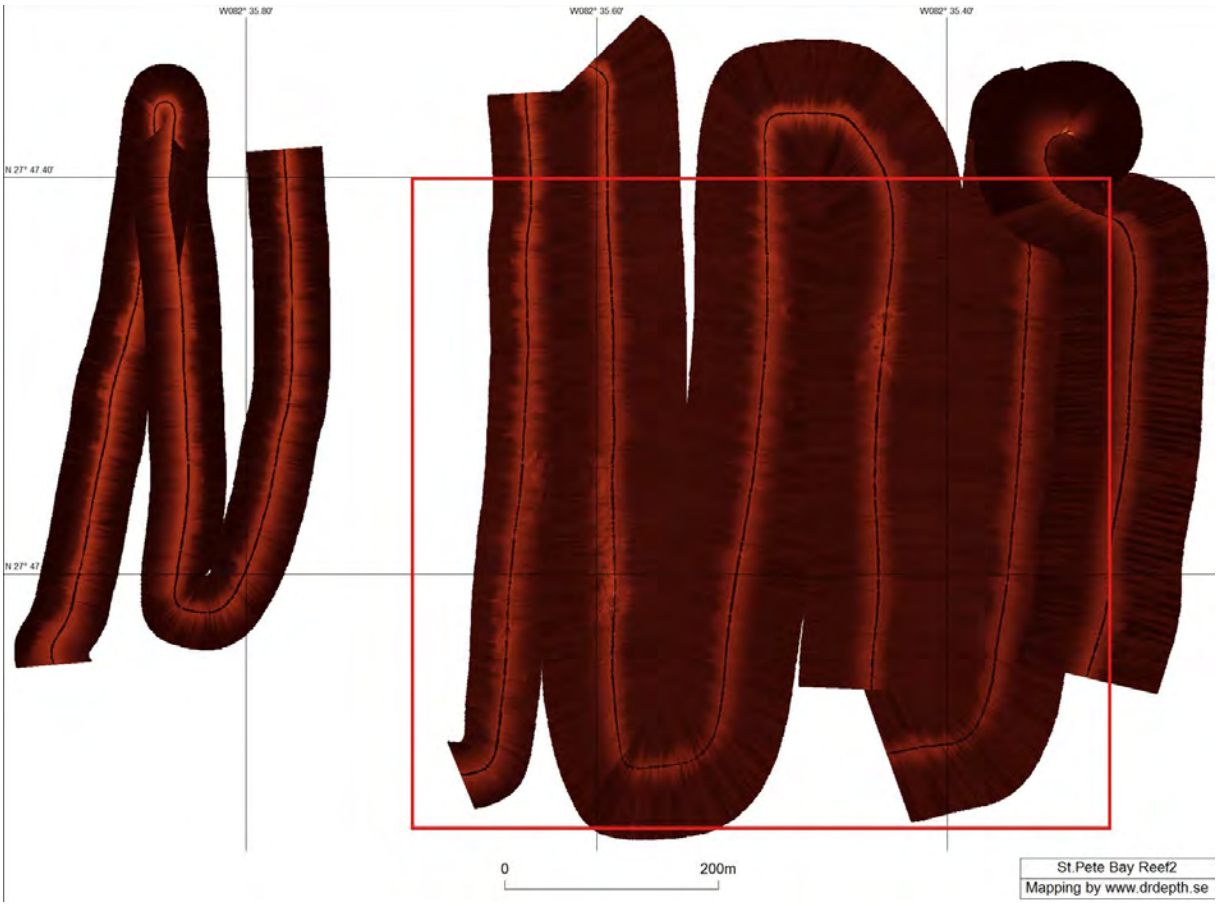


Fig 44. Selected St. Pete Bay Reef side scan area for estimation from side scan imagery.



Fig 45. St. Pete Bay Reef side scan area plotted in GIS with corresponding deployments. Total area was 3,918,588.46 sq ft. Materials estimated to cover about 5% of area mapped.

Sarasota County Deployments:

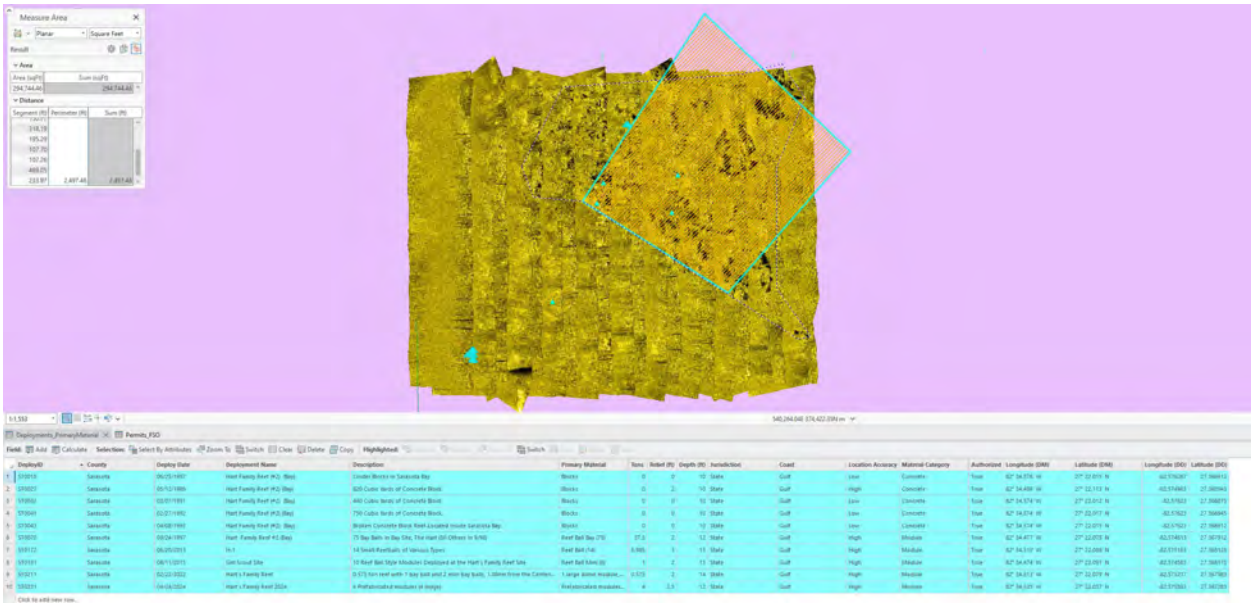


Fig 46. Side scan area for Hart Family Reef with corresponding deployments. Total permit area was 294,744.46 sq ft. Materials estimated to cover about 25% of permit area. There are 10 deployments in this area.

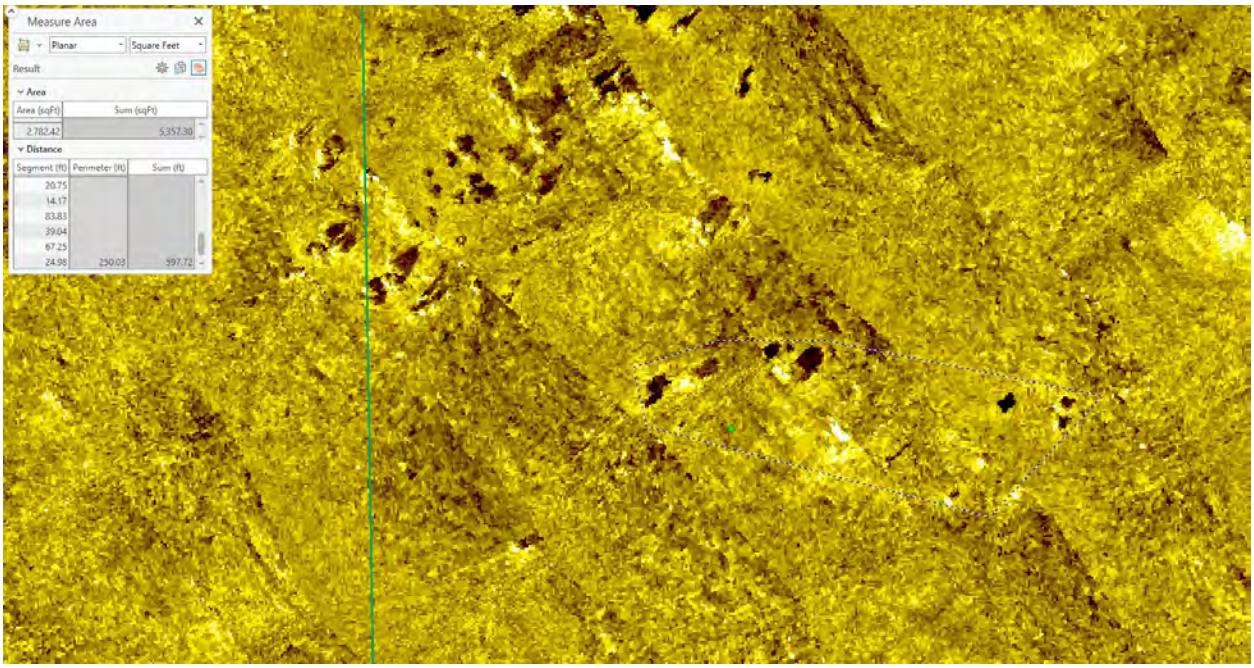


Fig 47. Deployment area estimated for ST0171by drawing polygon around materials, around 5,000 sq ft.